Mean Field Control, Mean Field Games and Reinforcement Learning

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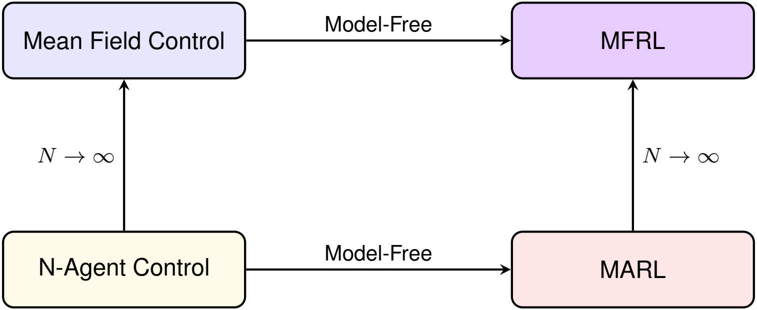
Outline

Motivations

Goal: Solve large-scale multi-agent problems.

Scalability:

- **Population Size**: via **Mean Field Approximation** ($N \to \infty$).
- **Environment Complexity**: via **(Deep) Reinforcement Learning** (Model-free).



Outline

Markov Decision Process (MDP)

An MDP is defined by a tuple (S, A, P, c, γ):

- **State Space** S: Set of possible situations.
- **Action Space** A: Set of possible decisions.
- **Transition Kernel** $P(x'|x, a)$: Probability of moving to x' given (x, a). Sometimes expressed with **transition function** F: $x' = F(x, a, \epsilon)$.
- **Cost Function** $c(x, a) \in \mathbb{R}$: Immediate cost after taking action a in state x.
- **Discount Factor** $\gamma \in [0, 1)$: Importance of future costs.

The Objective

Find a policy $\pi(a|x)$ that minimizes the **expected cumulative cost**:

$$J(\pi) = \mathbb{E}^{\pi}\left[\sum_{n=0}^{\infty} \gamma^n c(x_n, a_n)\right]$$

where $x_{n+1} \sim P(\cdot|x_n, a_n)$ and $a_n \sim \pi(\cdot|x_n)$.

Running Example: Cybersecurity – Single Agent

Consider a single server in an environment.

- **States**: $x \in \{DI, DS, UI, US\}$.
  - **D/U**: Defended / Undefended.
  - **I/S**: Infected / Susceptible.
- **Action**: $a \in [0, 1]$ (level of protection).
- **Dynamics**: $x_{n+1} = F(x_n, a_n, \epsilon_n)$, where
  - a_n influences the probability of switching between levels of defense (D/U).
  - ϵ_n represents the risk of infection.
- **One-stage Cost**: $c(x, a) = k_D \mathbb{I}_{x \in \{DI, DS\}} + k_I \mathbb{I}_{x \in \{DI, UI\}}$.

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Remarks:

- For simplicity, the cost depends only on the state.
- Here, the transition depends only on the agent's own state and action. Later on we will include interactions with other agents.

Dynamic Programming and Optimal Policy

- **State-Value Function**: $V^{\pi}(x) = \mathbb{E}^{\pi}\left[\sum_{k=0}^{\infty} \gamma^k c(x_{n+k}, a_{n+k}) \mid x_n = x\right]$.
- **Optimal Value**: $V^*(x) = \min_{\pi} V^{\pi}(x)$
- **Dynamic Programming**:

$$V^*(x) = \min_{a} \left\{c(x,a) + \gamma \mathbb{E}[V^*(x')]\right\}.$$

- **Computation**: Value Iteration, . . .

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Limitations:

- ✗ (L1): From V^* only, we cannot recover an optimal policy.
- ✗ (L2): Requires perfect knowledge of P and c.
- ✗ (L3): Fails for large/continuous spaces due to *Curse of Dimensionality*.
- ✗ (L4): Limited to one/a few agents.

Q-Function

State-Action Value Function

- **State-Action Value (Q-function)**:

$$Q^\pi(x,a) = \mathbb{E}^\pi\left[\sum_{k=0}^{\infty} \gamma^k c(x_{n+k}, a_{n+k}) \mid x_n = x, a_n = a\right]$$

- **Optimal Q-function**: $Q^*(x,a) = \min_\pi Q^\pi(x,a)$.

Connection with V-function: $V^*(x) = \min_a Q^*(x,a)$

Bellman Equation

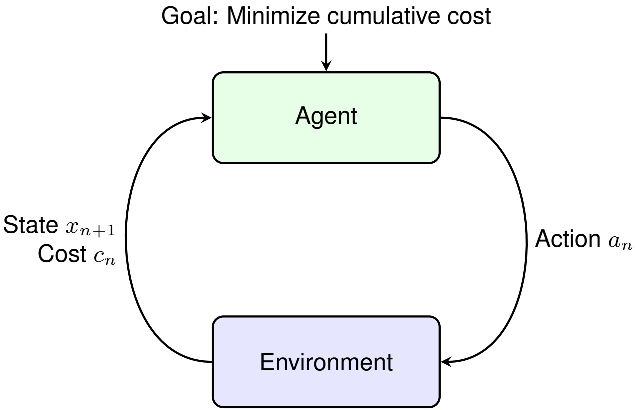
$$Q^*(x,a) = c(x,a) + \gamma \mathbb{E}_{x' \sim P(\cdot|x,a)}[\min_{a'} Q^*(x', a')]$$

Optimal Policy Recovery: If the optimal Q-function is known, an optimal policy can be obtained by greedy decisions:

$$\pi^*(x) \in \arg\min_{a \in A} Q^*(x,a) = \arg\min_{a \in A} \left\{c(x,a) + \gamma \mathbb{E}_{x' \sim P(\cdot|x,a)}[V^*(x')]\right\}$$

✓ Answer to (L1)

The Reinforcement Learning Paradigm



- **Agent**: The learner/decision maker.
- **Environment**: Everything the agent interacts with.
- **State** x_n: Current situation of the environment.
- **Action** a_n: Choice made by the agent.
- **Cost** $c_n = c(x_n, a_n)$: One-step (running) cost.

Model-Free Learning: TD Learning

- **Challenge**: Dynamics $P(x'|x,a)$ and costs $c(x,a)$ are often unknown
- **Goal:** Design *model-free* algorithms to learn Q^* or π^*.

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- **Temporal Difference (TD)**: Update estimates based on individual transitions (x_n, a_n, c_n, x_{n+1}).

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Q-Learning Update (Watkins, 1989)

The estimate $Q(x_n, a_n)$ is updated towards the **TD Target** $c_n + \gamma \min_{a'} Q(x_{n+1}, a')$:

$$Q(x_n, a_n) \leftarrow (1-\eta_n) Q(x_n, a_n) + \eta_n \left(c_n + \gamma \min_{a'} Q(x_{n+1}, a') \right)$$

where $\eta_n \in (0,1]$ is the learning rate.

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- Converges to Q^* if all pairs (x,a) are visited infinitely often.
- Trade-off: **Exploration** (try new a) vs **Exploitation** (choose $\min_a Q$).
- Often: **Exploring starts** assumption.

✓ Answer to (L2)

Motivations for Deep RL

- **Scalability Issue**: Tabular RL (storing Q in a table) fails when spaces S and/or A are very large or continuous (*Curse of Dimensionality*).
- We want to approximate Q as a function $S \times A \to \mathbb{R}$.
- **Solution**: Approximate $Q(x, a) \approx Q_\theta(x, a)$ using a neural network with parameters θ.

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DQN (Deep Q-Network) **(Mnih et al., 2015)**

Approximate Q^* using a neural net Q_θ by minimizing the loss:

$$\mathcal{L}(\theta) = \mathbb{E}_{(x,a,c,x')\sim\mathcal{D}}\left[\left(\underbrace{c + \gamma \min_{a'} Q_{\theta^-}(x',a')}_{\text{Target } y} - Q_\theta(x,a)\right)^2\right]$$

- **Experience Replay**: Break correlations by sampling from a buffer $\mathcal{D}$.
- **Target Networks**: Target parameters θ^- are fixed between updates.

✓ Answer to (L3)

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- **Experience Replay**: Break correlations by sampling from a buffer $\mathcal{D}$.
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✓ Answer to (L3) but requires *finite* action space.

- **Idea**: Parameterize the policy $\pi_\theta(a|x)$ and optimize θ directly to minimize $J(\theta)$.

Policy Gradient Theorem (Sutton et al., 1999)

The gradient $\nabla_\theta J(\theta)$ can be formulated as:

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim\pi_\theta}\left[\sum_{n=0}^{\infty} \nabla_\theta \ln \pi_\theta(a_n|x_n) \cdot \left(\sum_{k=n}^{\infty} \gamma^{k-n} c_k\right)\right]$$

Policy Gradient Methods

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- **Problem**: How to estimate the terms in the gradient?
- **REINFORCE**: Uses sample returns (trajectory-based).

✓ Answer to (L3) but suffers from high variance.

DDPG: Deep Deterministic Policy Gradient

Idea: Replace sum of future rewards by a learned Q-function.

Actor-Critic Architecture

- **Actor**: Represents the policy $\pi_\phi(x)$, mapping states to actions.
- **Critic**: Approx. the value function $Q_\psi(x, a)$, evaluating chosen actions.

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DDPG (Lillicrap et al., 2016)

An off-policy algorithm designed for **continuous action spaces**:

- Uses a deterministic actor $a = \pi_\phi(x)$.
- Updates ϕ by following the gradient through the critic:

$$\nabla_\phi J \approx \mathbb{E}\left[\nabla_a Q_\psi(x, a)|_{a=\mu_\phi(x)} \nabla_\phi \pi_\phi(x)\right]$$

Note: For *stochastic* policies, see **SAC**, **PPO**, ...

✓ Answer to (L3): Scalable and handles continuous S and A.

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✓ Answer to (L3): Scalable and handles continuous S and A.

✗ *How about (L4)?* MFRL.

Outline

N-Agent System

Consider a system with $N \geq 1$ agents (finite population).

- **States**: $\underline{x}_n = (x_n^1, \dots, x_n^N) \in S^1 \times \cdots \times S^N$.
- **Action Profile**: $\underline{a}_n = (a_n^1, \dots, a_n^N) \in A^1 \times \cdots \times A^N$.
- **Policies**: Each agent i follows a policy $a_n^i = \pi_n^i(\underline{x}_n)$.
- **Policy Vector**: $\underline{\pi} = (\pi^1, \dots, \pi^N)$.
- **Objective**: Each agent i aims at minimizing:

$$J^i(\underline{\pi}) = \mathbb{E}\left[\sum_{n=0}^{\infty} \gamma^n f^i(\underline{x}_n, \underline{a}_n)\right]$$

Dynamics and Noise

- ϵ_n^i: **Idiosyncratic noise** (independent for each agent).
- ϵ_n^0: **Common noise** (affects everyone simultaneously).
- Private state x^i evolves according to:

$$x_{n+1}^i = F^i(\underline{x}_n, \underline{a}_n, \epsilon_n^i, \epsilon_n^0)$$

Now consider N servers connected in a network.

- **State Vector**: $\underline{x}_n = (x_n^1, \ldots, x_n^N)$.
- **Coupled Dynamics**: For server i, the evolution is:

  $$x_{n+1}^i = F(x_n^i, a_n^i, \underline{x}_n, \epsilon_n^i)$$

  where the **infection risk** for server i depends on the health status of **other** servers in the network (e.g., malware spread).
- **One-stage Cost**: $c^i(x^i, a^i) = k_D \mathbb{I}_{x^i \in \{DI, DS\}} + k_I \mathbb{I}_{x^i \in \{DI, UI\}}$.

Symmetry and Homogeneity

To study the limit $N \to \infty$, we assume specific structures:

1. Symmetric Interaction (Anonymity)

The interaction depends only on the **empirical distribution** of states and actions:

- **Empirical State-Action Measure**: $\hat{\nu}_n^N = \frac{1}{N}\sum_{i=1}^N \delta_{(x_n^i, a_n^i)}$.

Then for all i,

$$f^i(\underline{x}_n, \underline{a}_n) = f^i(x_n^i, a_n^i, \hat{\nu}_n^N), \qquad F^i(\underline{x}_n, \underline{a}_n, \epsilon_n^i, \epsilon_n^0) = F^i(x_n^i, a_n^i, \hat{\nu}_n^N, \epsilon_n^i, \epsilon_n^0).$$

2. Homogeneity (Identical Agents)

All agents share the same spaces S, A, and the same cost and dynamics functions:

$$f^i = f, \qquad F^i = F, \qquad \text{for all } i \in \{1, \dots, N\}.$$

This structure ensures that agents are *interchangeable*.

The Optimization Landscape

Two major forms of multi-agent optimization:

1. Cooperation: Social Optimum

Agents cooperate to minimize the aggregate (average) cost:

$$\min_{\underline{\pi}} \tilde{J}(\underline{\pi}) = \min_{\underline{\pi}} \frac{1}{N} \sum_{i=1}^{N} J^i(\underline{\pi})$$

2. Competition: Nash Equilibrium

A profile $\underline{\pi}^*$ is a Nash equilibrium if no agent i can deviate alone to reduce their own cost:

$$J^i(\underline{\pi}^{*-i}, \pi') \geq J^i(\underline{\pi}^*) \quad \text{for any admissible } \pi'.$$

Focus: We will focus on the **cooperative** case.

Running Example: Cybersecurity – Social Optimum

By assuming symmetry and large N, the problem structure simplifies:

- **Symmetric Interaction**: The infection risk for a server depends only on the **proportion** of infected servers in the population:

$$\hat{\mu}_n^N = \frac{1}{N}\sum_{j=1}^{N} \delta_{x_n^j}$$

- **Cooperative Mean Field Setting**: The one-stage social cost is the integral over states and actions:

$$\frac{1}{N}\sum_{i=1}^{N} c(x_n^i, a_n^i) = k_D \hat{\mu}_n^N(\{DI, DS\}) + k_I \hat{\mu}_n^N(\{DI, UI\})$$

The total discounted cost to minimize is:

$$\mathbb{E}\left[\sum_{n=0}^{\infty} \gamma^n \left[k_D \hat{\mu}_n^N(\{DI, DS\}) + k_I \hat{\mu}_n^N(\{DI, UI\})\right]\right]$$

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Note: We need $\mathbb{E}$ because $\hat{\mu}_n^N$ is stochastic. It becomes deterministic in the limit $N \to +\infty$ unless there is common noise.

Challenges for Large N

Solving the N-agent cooperative problem directly is hard:

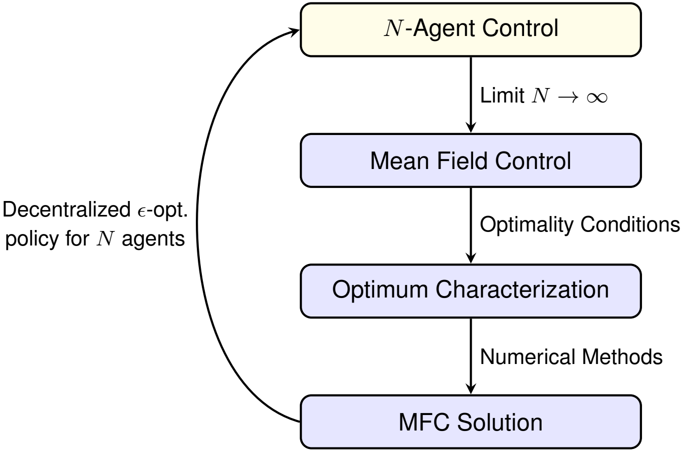
- **Curse of Dimensionality**: The state space $S^1 \times \cdots \times S^N$ grows exponentially with N.
- **Policy Complexity**: Storing and computing N different policies $\underline{\pi} = (\pi^1, \ldots, \pi^N)$ is prohibitive.

The Mean Field Idea

Under the **Symmetry** and **Homogeneity** assumptions:

- The problem simplifies as $N \to \infty$.
- We obtain a **Mean Field Control (MFC)** problem.
- We look for a *single* representative policy π instead of N.

The Mean Field Control Paradigm



Mean Field Control: Discrete-Time Model

In the limit $N \to \infty$, a conditional form of the **propagation of chaos** leads to conditional independence of individual dynamics.

Representative Agent Dynamics

The state of a representative agent x_n evolves according to:

$$x_{n+1} = F(x_n, a_n, \mathbb{P}^0_{(x_n,a_n)}, \varepsilon_{n+1}, \varepsilon^0_{n+1}), \quad n \geq 0,$$

where $\mathbb{P}^0_{(x_n,a_n)}$ stands for the conditional law of the state-action couple (x_n, a_n) given the common noise $(\varepsilon^0_n)_{n\geq 1}$.

MFC Optimization Problem

The objective in this limiting regime is:

$$\inf_\pi J(\pi), \qquad J(\pi) = \mathbb{E}\left[\sum_{n=0}^{\infty} \gamma^n f\left(x_n, a_n, \mathbb{P}^0_{(x_n,a_n)}\right)\right].$$

This intuition is substantiated by rigorous mathematical consistency results.

Common Randomization and Sources of Stochasticity

Analogously to action randomization in standard MDPs, the central planner may use **common randomization**.

Common Randomization

- **Level-1:** The planner samples a policy π_n from a distribution using a common random variable (θ_n^0). This π_n is shared by all agents.
- **Level-0:** Each agent samples an action a_n from $\pi_n(x_n)$ using individual randomization.
- The conditional distribution $\mathbb{P}^0$ is then conditioned on **both** the common noise (ε_n^0) and the common randomization (θ_n^0).

Advantage: Allows exploration at the population-level.

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Summary: Four Types of Randomness

	Individual / Idiosyncratic	Common / Global
Noise	Agent-specific shocks ε^i	External environment ε^0
Randomization	Randomized local policy	Shared global policy sampling

For a policy π, the MFC value function J^π corresponds to an MDP on the space of measures. The **lifted MFMDP** is a six-tuple $(\bar{S}, \bar{A}, \bar{\Gamma}, P, \bar{f}, \gamma)$:

- **State Space**: $\bar{S} := \mathcal{P}(S)$. Generic element μ.
- **Action Space**: $\bar{A} := \mathcal{P}(S \times A)$. Generic element $\bar{a}$.
- **Graph**: $\bar{\Gamma} := \{(\mu, \bar{a}) \in \bar{S} \times \bar{A} \mid \bar{a} \in \bar{U}(\mu)\}$, where the **Control Constraint** is $\bar{U}(\mu) := \{\bar{a} \in \bar{A} \mid \mathrm{pr}_1(\bar{a}) = \mu\}$.
- **Transition Kernel**: $\bar{P} : \bar{\Gamma} \to \mathcal{P}(\bar{S})$, a measurable function.
- **One-stage Cost**: $\bar{f} : \bar{\Gamma} \to \mathbb{R}$, bounded and measurable.
- **Discount factor**: $\gamma \in (0, 1)$.

MFMDP Lifted from MFC Model

The MFMDP is **consistent** with the MFC model $(S, A, E, E^0, F, f, \gamma)$ if:

- **Transition Kernel**:

$$\bar{P}(\mu, \bar{a})(d\mu') = \left(\nu^0 \circ \bar{F}(\mu, \bar{a}, \cdot)^{-1}\right)(d\mu')$$

  where the lifted system function $\bar{F}$ is defined by:

$$\bar{F}(\mu, \bar{a}, e^0) = (\bar{a} \otimes \nu) \circ F(\cdot, \cdot, \bar{a}, \cdot, e^0)^{-1}$$

- **Lifted Cost Function**:

$$\bar{f}(\mu, \bar{a}) = \int_{S \times A} f(x, a, \bar{a})\, \bar{a}(dx, da)$$

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- **Lifted Cost Function**:

$$\bar{f}(\mu, \bar{a}) = \int_{S \times A} f(x, a, \bar{a})\, \bar{a}(dx, da)$$

Optimal value function: $\bar{J}^*$; satisfies Dynamic Programming Principle.

Running Example: Cybersecurity – MFMDP

At the mean field level:

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$$\mu_n = \mathbb{P}^0_{x_n}$$

- **Cooperative Mean Field Setting**: The one-stage social cost is the integral over states and actions:

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Note: We need $\mathbb{E}$ if there is common noise.

Two main frameworks when dealing with large populations:

- **Mean Field Control (MFC)**: Social optimum
    - **Optimality**: Find a central policy π^* to minimize the average cost.
    - **Characteristic**: No incentive for joint deviations.
    - **Mathematical nature**: An **optimization** problem.

From Control to Games

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  - **Optimality**: For a given mean field flow μ, each agent finds a best response $\hat{\pi}$.
  - **Consistency**: The resulting mean field flow from all agents using $\hat{\pi}$ must be exactly μ.
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  - **Mathematical nature**: A **fixed point** problem over the mean field μ.

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Focus: The tutorial is mainly focused on MF Control (cooperative), but we will present RL algorithms for MF Games (non-cooperative) later on.

Mean Field Games (MFG): Individual Cost

Individual Cost in MFG: In contrast to MFC where $J(\pi)$ is the average population cost, the cost for a representative agent in an MFG scenario depends on their own policy π and the population's behavior.

- **As a function of a mean field flow**: $J(\pi; \mu)$, where $\mu = (\mu_n)_{n \geq 0}$ is the population's state distribution sequence over time.

MFG Individual Cost Function

Given a population flow μ, the expected cumulative cost for a representative agent using policy π is:

$$J(\pi; \mu) = \mathbb{E}\left[\sum_{n=0}^{\infty} \gamma^n f\left(x_n, a_n, \mu_n\right)\right]$$

where individual dynamics are determined by $a_n \sim \pi(\cdot|x_n)$ and the environment F.

Mean Field Games (MFG): Population Policy

Alternatively, we can define the cost assuming the population uses a shared policy π'.

- Individual cost rewritten as $J(\pi; \pi')$, where π is the agent's policy and π' is the population's policy.
- This is evaluated by setting $\mu = \mu^{\pi'}$, the distribution flow generated by π'.

Population Distribution Evolution

The sequence $\mu^{\pi'} = (\mu_n^{\pi'})_{n \geq 0}$ follows a Markov chain-type evolution: Assuming no common noise,

$$\mu_{n+1}^{\pi'}(x') = \sum_{x \in S} \sum_{a \in A} P(x'|x,a)\pi'(a|x)\mu_n^{\pi'}(x)$$

starting from an initial distribution $\mu_0^{\pi'} = \mu_0$. Thus, we define:

$$J(\pi; \pi') := J(\pi; \mu^{\pi'})$$

MFG Individual Cost Reformulation

With the above notation, we define:

$$J(\pi; \pi') := J(\pi; \mu^{\pi'})$$

Mean Field Nash Equilibrium (MFNE) Definition

For a fixed population behavior μ, the agent aims to find a **best response** policy $\hat{\pi}$ by solving a standard Markov Decision Process: $\hat{\pi} \in \arg\min_\pi J(\pi; \mu)$.

A Mean Field Nash Equilibrium (MFNE) characterizes a state where no agent has an incentive to unilaterally deviate. It can be formulated in two equivalent ways:

Definition 1 (Policy-based)

A policy $\hat{\pi}$ is a Mean Field Nash Equilibrium if it is a best response against itself:

$$\hat{\pi} \in \arg\min_{\pi} J(\pi; \hat{\pi})$$

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Definition 2 (Policy-Distribution pairs)

A pair $(\hat{\pi}, \hat{\mu})$ is a Mean Field Nash Equilibrium if it satisfies exactly two conditions:

1. **Optimality (Best Response)**: $\hat{\pi}$ is optimal given the fixed population $\hat{\mu}$:

$$\hat{\pi} \in \arg\min_{\pi} J(\pi; \hat{\mu})$$

2. **Consistency (Mean Field)**: The distribution flow $\hat{\mu}$ is exactly the one generated when the whole population uses policy $\hat{\pi}$:

$$\hat{\mu}_n = \mathbb{P}_{x_n^{\hat{\pi}}} \quad \text{for all } n \geq 0$$

Further Reading: Foundations and Surveys

- **MFG foundations**: Lasry–Lions [LL07a]; Nash certainty equivalence [HMC06]; monographs and lecture notes [BFY+13, CD18a, CD18b, ACD+20].
- **Mean field control and MFMDP**: population MDP viewpoint [GG11, GGB12]; probabilistic MFC and McKean–Vlasov control [CLT23a, CD18a, MP22]; dynamic programming for MFMDPs [CLT23b, MP23].
- **Single-agent and multi-agent RL background**: stochastic control and MDPs [BS96, Put14]; MARL surveys [CTE+22a, ZYB21].
- **Numerics and learning surveys**: numerical methods for MFG/MFC [Lau21]; machine learning for stochastic control/games [HL22]; learning in MFGs [LPP+].

Further Reading: Applications

- **Crowds, flocking, mobility**: crowd motion and routing [ALL19, PPL$^+$20, CLP$^+$22]; flocking [NCM11, PLP$^+$21]; urban mobility and traffic [DXS25, MCD$^+$24].
- **Economics and finance**: early application surveys [GLL11]; systemic risk and finance/economics [CS15, Car21, BVTR23]; social networks [NM18].
- **Energy, security, communications**: energy systems [ATM20, EHMP19, GB18]; security and wireless communications [MVL12, SBS$^+$15, YLS$^+$17, KM18].
- **Engineering and autonomous systems**: autonomous vehicles and traffic control [HDDC17, HDDC19, TNPJ20]; safe mean-field MARL [JPJ$^+$24].

Outline

This part is mostly based on:

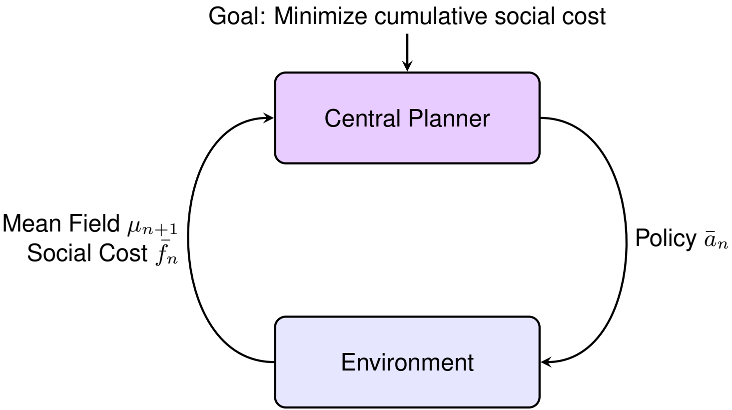
Carmona, Lauriere, Tan: AAP 2023

and lecture notes

Carmona & Lauriere: Mean Field Reinforcement Learning
(arXiv:2607.01525)

Outline

The Mean Field RL Paradigm



- **Central Planner**: Samples a policy shared by the population.
- **Environment**: Global dynamics of the population.
- **Mean Field** μ_n: State of the population (distribution).
- **Policy** $\bar{a}_n$: Corresponds to a Level-1 action in the MFMDP.
- **Social Cost** $\bar{f}_n = \bar{f}(\mu_n, \bar{a}_n)$: Aggregate population cost.

Controls for Finite State and Action Spaces

- Assume individual spaces S and A are finite.
- $\mathcal{P}(S)$ and $\mathcal{P}(A)$ are simplices.
- Marginal distribution constraints:

$$\sum_a \bar{a}(x, a) = \mu(x).$$

- Central Planner Actions:

$$\tilde{A} := \{\tilde{a} : S \to \mathcal{P}(A) \mid \tilde{a} \text{ Borel measurable}\}$$

- Joint distribution: $\mu \,\hat{\otimes}\, \tilde{a}$.
- Q-Function rewritten:

$$\tilde{Q}^*(\mu, \tilde{a}) := \bar{Q}^*(\mu, \mu \,\hat{\otimes}\, \tilde{a}).$$

Example: Cybersecurity Model Revisited

- **Scenario**: A central planner protects a large network of computers.
- **Individual States**: Each computer is in one of 4 states ($|S| = 4$):
  - **D**efended or **U**ndefended
  - **I**nfected or **S**usceptible
- **Dynamics**:
  - Computers recover from infection at fixed rates.
  - New infections occur via *external* hackers or *internal* spread.
  - **Mean Field Interaction**: Internal spread rates (e.g., $\beta\mu(\{DI\})$) depend on the percentage of infected neighbors.
- **Goal**: Minimize protection costs + infection penalties across the population.

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Mathematically:

- **State Set**: $S = \{DI, DS, UI, US\}$.
- **Transitions**: The intensity of transitioning from $DS \to DI$ is:

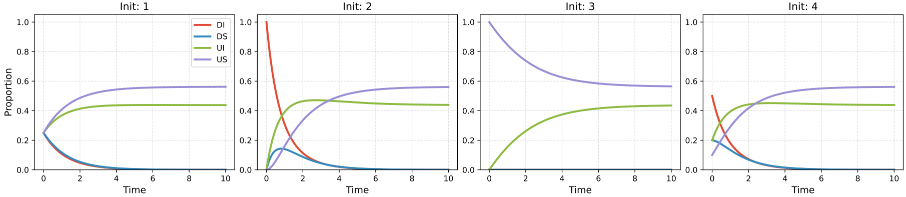
$$P^{\mu,a}_{DS\to DI} = v_H q^D_{inf} + \beta_{DD}\mu(\{DI\}) + \beta_{UD}\mu(\{UI\})$$

- **Total Cost**: The central planner pays for defense and infection:

$$\bar{f}(\mu, \bar{a}) = k_D\mu(\{DI, DS\}) + k_I\mu(\{DI, UI\})$$

Continuous-time benchmark:

- solution through forward-backward ODEs
- for each *given* initial distribution



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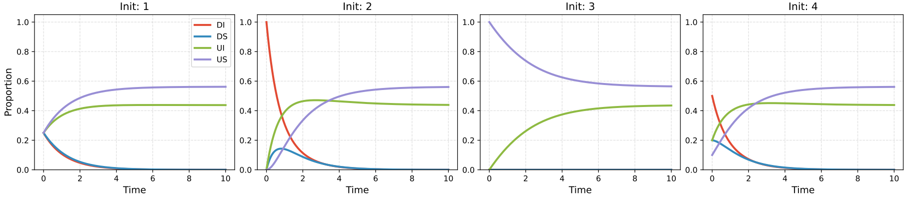

Goal: learn the optimal action

- for every individual state
- for *every* mean field distribution

Outline

Simplex Discretization and Tabular MFQ-Learning

- $\mathcal{P}(S)$ and $\mathcal{P}(A)$ are infinite continuous simplices.
- We want to start with a **simple tabular RL method** for finite spaces.
- **Idea**: Discretize $\mathcal{P}(S)$ and $\mathcal{P}(A)$ onto finite grids: $\check{\mathfrak{S}} \subset \mathcal{P}(S)$ and $\check{\mathfrak{A}} \subset \mathcal{P}(A)$.

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- **Resolution Parameters**:
  - $\varepsilon_{\mathfrak{S}}$: State grid mesh size (e.g., $1/N_m$ where N_m is the mesh divisions).
  - $\varepsilon_{\mathfrak{A}}$: Action grid mesh size.
- **Cost Function**: $\tilde{f}(\mu, \tilde{a}) = \bar{f}(\mu, \mu \hat{\otimes} \tilde{a})$ be the planner's cost given $\tilde{a} : S \to \mathcal{P}(A)$.
- **Projection**: At each step, project onto the grid:

$$\check{\mu}_{n+1} = \text{proj}_{\check{\mathfrak{S}}} \circ \bar{F}(\check{\mu}_n, \check{\mu}_n \hat{\otimes} \check{a}, \varepsilon^0_{n+1})$$

(Alternatively: project only when updating the Q-table)

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Cybersecurity example:

- **State Simplex**: $\mathfrak{S} = \{(\mu^{(1)}, \dots, \mu^{(4)}) \in [0,1]^4 \mid \sum_i \mu^{(i)} = 1\}$.
- **Discretized Grid**: $\check{\mathfrak{S}} = \{(\mu^{(i)}) \in \{0, 1/N_m, \dots, 1\}^4 \mid \sum_i \mu^{(i)} = 1\}$.

The Curse of Dimensionality: $|\check{\mathfrak{S}}| = \binom{N_m+|S|-1}{|S|-1}$, poly. in N_m but *expo.* in $|S|$:

$N_m \setminus \lvert S \rvert$	2	3	4	5
10	11	66	286	1,001
20	21	231	1,771	10,626
50	51	1,326	**23,426**	316,251

The MFQ-Learning Algorithm

Algorithm: Tabular MFQ-Learning with Discretization

Input: Grids $\check{\mathfrak{S}}$, $\check{A}$, episodes N_{total}, length N_{epi}, learning rate η.
Output: Trained Q-table $\check{Q}$.

Initialize table $\check{Q}_0$ on $\check{\mathfrak{S}} \times \check{A}$
for $m = 1, \ldots, N_{total}$ **do**
    Sample initial distribution $\mu_0 \in \mathfrak{S}$ and set $\check{\mu}_0 = \text{proj}_{\check{\mathfrak{S}}}(\mu_0)$
    for $n = 0, \ldots, N_{epi} - 1$ **do**
        Select action $\check{a}_n$ from $\check{\mu}_n$ (e.g., using ϵ-greedy)
        Execute $\check{a}_n$, observe cost $\tilde{f}(\check{\mu}_n, \check{a}_n)$ and new state:
            $\check{\mu}_{n+1} = \text{proj}_{\check{\mathfrak{S}}} \circ \bar{F}(\check{\mu}_n, \check{\mu}_n \hat{\otimes} \check{a}_n, \varepsilon^0_{n+1})$
        Update:
            $\check{Q}(\check{\mu}_n, \check{a}_n) \leftarrow (1 - \eta_k)\check{Q}(\check{\mu}_n, \check{a}_n) + \eta_k \left(\tilde{f}(\check{\mu}_n, \check{a}_n) + \gamma \min_{a'} \check{Q}(\check{\mu}_{n+1}, a') \right)$
return $\check{Q}$

Q-Learning with Mixed vs. Pure Controls

Theorem (MFQ-Learning Convergence (Informal))

The learned table $\check{Q}$ converges to the true infinite-population Q-function, up to:

- *tabular Q-learning error*
- *discretization error.*

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Extra: Learned (greedy) policy converges.

Two scenarios:

- **Mixed Controls (Level-0 Randomization)**:
  - Representative agent uses randomized actions: $\tilde{a} : S \to \mathcal{P}(A)$.
  - $\tilde{A} \approx \mathcal{P}(A)^{|S|}$ requires discretizing both state and action simplices.
  - Error bound ε' depends on both grid resolutions $\varepsilon_{\mathfrak{S}}$ and $\varepsilon_{\mathfrak{A}}$.
- **Pure Controls (Deterministic Level-0)**:
  - In some MFC problems, optimal policies are non-randomized at level-0.
  - Representative agent uses deterministic actions: $\tilde{a} : S \to A$.
  - Action space $\tilde{A} \approx A^S$ is finite; no discretization of $\mathcal{P}(A)$ needed.
  - Computationally cheaper: updating a table of size $|\check{\mathfrak{S}}| \times |A|^{|S|}$.

Assumptions (Informal)

Regularity and Exploration Assumptions

- **Lipschitz Data**: Cost $\bar{f}$ and dynamics $\bar{F}$ are Lipschitz in $(\mu, \tilde{a})$.
- **Value Function**: Optimal $\mu \mapsto \bar{J}^*(\mu)$ is Lipschitz continuous.
- **Covering Time**: $(\check{\mu}_n, \check{a}_n)$ covers the grid $\check{\mathfrak{S}} \times \check{A}$.
- **Learning Rates**: $\eta_n = 1/(1 + C(n, \check{\mu}, \check{a}))$, with $C(n, \check{\mu}, \check{a}) =$ nb of visits.

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Theorem (Q-Table Convergence (Informal))

Let $\delta \in (0, 1)$ and $\varepsilon > 0$. If the total number of steps is of order

$$N_{\text{total}} \times N_{\text{epi}} = \Omega\left(\frac{\ln(1/\delta)^5}{\varepsilon^2}\right),$$

then with probability $1 - \delta$, for all $(\mu, \tilde{a}) \in \mathfrak{S} \times \tilde{\mathcal{A}}$,

$$\left|\check{Q}_{N_{\text{total}}}\left(\mathrm{Proj}_{\check{\mathfrak{S}}}(\mu), \mathrm{Proj}_{\check{\mathcal{A}}}(\tilde{a})\right) - \bar{Q}^*(\mu, \mu \hat{\otimes} \tilde{a})\right| \leq \varepsilon',$$

where $\varepsilon' = \mathcal{O}\big(\varepsilon + \big(\frac{1}{1-\gamma} + 1\big)\varepsilon_{\mathfrak{S}} + \frac{1}{1-\gamma}\varepsilon_{\mathfrak{A}}\big)$.

Remarks

- **Covering time:**
  - $T_{cov} = |\check{\mathfrak{S}} \times \check{\mathcal{A}}|$ if full sweep
  - In practice: follow trajectory
  - Enough to assume that the covering is smaller than L with probability $1/2$.

- **Regularity:** Extension to the case where $\bar{J}^*$ is only p-Hölder continuous.

Detailed Assumptions

Assumption: Regularity of the data

$\tilde{f}$ is bounded and Lipschitz continuous with respect to $(\mu, \tilde{a})$ with constant $L_{\tilde{f}}$, namely for every $(\mu, \tilde{a}), (\mu', \tilde{a}') \in \mathfrak{S} \times \tilde{\mathcal{A}}$, we have

$$|\tilde{f}(\mu, \tilde{a}) - \tilde{f}(\mu', \tilde{a}')| \leq L_{\tilde{f}} \left(\|\mu - \mu'\|_{d_{\mathfrak{S}}} + d_{\tilde{\mathcal{A}}}(\tilde{a}, \tilde{a}') \right) \qquad and \qquad \tilde{f}(\mu, \tilde{a}) \leq L_{\tilde{f}}.$$

Also, $\bar{F}$ is Lipschitz continuous with respect to μ and $\tilde{a}$ with constant $L_{\bar{F}}$ in expectation over the randomness of the common noise, namely for every $(\mu, \tilde{a}), (\mu', \tilde{a}') \in \mathfrak{S} \times \tilde{\mathcal{A}}$:

$$\mathbb{E}_{\varepsilon^0} \left[\|\bar{F}(\mu, \mu \hat{\otimes} \tilde{a}, \varepsilon^0) - \bar{F}(\mu', \mu' \hat{\otimes} \tilde{a}', \varepsilon^0)\|_{d_{\mathfrak{S}}} \right] \leq L_{\bar{F}} \left(\|\mu - \mu'\|_{d_{\mathfrak{S}}} + d_{\tilde{\mathcal{A}}}(\tilde{a}, \tilde{a}') \right).$$

Assumption: Regularity of the value function

$\bar{J}^*$ is Lipschitz continuous w.r.t. μ with constant $L_{\bar{J}^*}$.

Assumption: Covering time

There exists a finite T_{cov} such that for every n_0, $\check{\mathfrak{S}} \times \check{\mathcal{A}} \subseteq (\check{\mu}_n, \check{a}_n)_{n=n_0,\ldots,n_0+T_{cov}}$.

Assumption: Learning rate

There exists $\kappa \in (1/2, 1)$ such that for every $(\check{\mu}, \check{a}) \in \check{\mathfrak{S}} \times \check{\mathcal{A}}$, $\eta_n := \eta_n(\check{\mu}, \check{a}) = 1/\big(1 + C(n, \check{\mu}, \check{a})\big)^\kappa$ for each $n \geq 0$, where $C(n, \check{\mu}, \check{a})$ is the number of times up to n that the pair $(\check{\mu}, \check{a})$ has been visited in the MFQ-Learning Algorithm.

Notations

- $\check{J}_{bound}$: upper bound on the optimal value of the projected MFC problem.

- $\mathrm{Proj}_{\check{\mathfrak{S}}} : \mathfrak{S} \to \check{\mathfrak{S}}$ and $\mathrm{Proj}_{\check{\mathcal{A}}} : \tilde{\mathcal{A}} \to \check{\mathcal{A}}$: projection operators such that

$$(\mathrm{Proj}_{\check{\mathfrak{S}}}(\mu), \mathrm{Proj}_{\check{\mathcal{A}}}(\tilde{a})) := (\check{\mu}, \check{a})$$

  for every $(\mu, \tilde{a}) \in \mathfrak{S} \times \tilde{\mathcal{A}}$ where $(\check{\mu}, \check{a})$ is the closest point (or one of the closest points, in case of equality) in $\check{\mathfrak{S}} \times \check{\mathcal{A}}$ with respect to $d_{\mathfrak{S}}$ and $d_{\tilde{\mathcal{A}}}$.

- Based on simplex discretizations:

$$\|\mu - \check{\mu}\|_{d_{\mathfrak{S}}} \leq \varepsilon_{\mathfrak{S}}, \qquad d_{\tilde{\mathcal{A}}}(\tilde{a}, \check{a}) \leq \varepsilon_{\mathfrak{A}}.$$

- $\beta = (1-\gamma)/2$ the horizon of the MFMDP

- For $\delta \in (0,1)$, we let $T_{cov}(\delta) = \lceil T_{cov} \log_2(1/(2\delta)) \rceil$.

Theorem (Q-Table Convergence)

Let $\delta \in (0,1)$ and $\varepsilon > 0$. If the total number of steps is of order

$$\Omega\left(\left(\frac{(T_{cov}(\delta))^{1+3\kappa}\check{J}^2_{bound}\ \ln\left(|\check{\mathfrak{S}}|\,|\check{\mathfrak{A}}|^{|S|}\check{J}_{bound}/(2\delta\beta\varepsilon)\right)}{\beta^2\varepsilon^2}\right)^{\frac{1}{\kappa}} + \left(\frac{(T_{cov}(\delta))}{\beta}\ln\left(\frac{\check{J}_{bound}}{\varepsilon}\right)\right)^{\frac{1}{1-\kappa}}\right),$$

then with probability $1-\delta$, for all $(\mu, \tilde{a}) \in \mathfrak{S} \times \tilde{\mathcal{A}}$,

$$\left|\check{Q}_{N_{\text{total}}}\left(\mathrm{Proj}_{\check{\mathfrak{S}}}(\mu), \mathrm{Proj}_{\check{\mathcal{A}}}(\tilde{a})\right) - \bar{Q}^*(\mu, \mu \hat{\otimes} \tilde{a})\right| \leq \varepsilon',$$

where $\varepsilon' = \varepsilon + \left(\frac{\gamma}{1-\gamma}L_{\bar{J}^*} + L_{\tilde{f}} + \gamma L_{\bar{J}^*}L_{\bar{F}}\right)\varepsilon_{\mathfrak{S}} + \frac{1}{1-\gamma}\left(L_{\tilde{f}} + \gamma L_{\bar{J}^*}L_{\bar{F}}\right)\varepsilon_{\mathfrak{A}}$.

Sketch of Proof

We decompose the error using the triangle inequality into three manageable terms:

$$\begin{aligned}\left|\check{Q}\left(\hat{\mu},\hat{a}\right)-\tilde{Q}^*\left(\mu,\tilde{a}\right)\right| \leq\ & \underbrace{|\check{Q}(\hat{\mu},\hat{a})-\check{Q}^*(\hat{\mu},\hat{a})|}_{\text{Step 1: Q-Learning Convergence}} \\ & +\underbrace{|\check{Q}^*(\hat{\mu},\hat{a})-\tilde{Q}^*(\hat{\mu},\hat{a})|}_{\text{Step 2: Discretization Stability}} \\ & +\underbrace{|\tilde{Q}^*(\hat{\mu},\hat{a})-\tilde{Q}^*(\mu,\tilde{a})|}_{\text{Step 3: Continuous Regularity}}\end{aligned}$$

where $(\hat{\mu},\hat{a}) = (\text{proj}_{\check{\mathcal{S}}}(\mu), \text{proj}_{\check{A}}(\tilde{a}))$.

1. **Step 1**: Standard asynchronous RL results (e.g. Even-Dar & Mansour):

$$\check{Q} \to \check{Q}^* \text{ as nb of steps } \to +\infty.$$

2. **Step 2**: Stability under projection.
3. **Step 3**: Continuous regularity: uses Lipschitz properties of $\tilde{f}$ and $\bar{J}^*$.

Value Function Convergence for Pure Controls

If agents use pure controls: no need to discretize $\mathcal{P}(\mathcal{A})$.

Theorem (Q-Table Convergence with Pure Level-0 Controls)

If the total steps is sufficiently large, the error bound simplifies:

$$\sup_{\mu, \tilde{a} \in A^S} \left| \check{Q}\left(\mathsf{proj}_{\check{\mathfrak{S}}}(\mu), \tilde{a}\right) - \tilde{Q}^*(\mu, \delta_{\tilde{a}}) \right| \leq \varepsilon',$$

where:

$$\varepsilon' = \mathcal{O}\left(\varepsilon + \left(\frac{1}{1-\gamma} + 1\right) \varepsilon_{\mathfrak{S}}\right)$$

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Recovering a control from a value function:

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We make use of an extra assumption.

Assumption: Action Gap

There exists $K_A > 0$ s.t. the optimal action is better than the second-best by K_A:

$$\tilde{Q}^*(\check{\mu}, \tilde{a}^*) - \check{Q}^*(\check{\mu}, \tilde{a}) \geq K_A, \quad \forall \tilde{a} \neq \tilde{a}^*$$

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Corollary: Policy Convergence

Under the above assumption, then:

$$\left\| \mathrm{softmin}_\tau \big(\check{Q}(\check{\mu}, \cdot)\big) - \mathrm{argmine}\big(\tilde{Q}^*(\check{\mu}, \cdot)\big) \right\|_2 \leq \tau \varepsilon' \sqrt{|\tilde{A}|} + 2e^{-\tau K_A} |\tilde{A}|$$

Notation: $\mathrm{softmin}_\tau : \mathbb{R}^m \to \mathbb{R}^m$ by $\mathrm{softmin}_\tau(x) = (e^{-\tau x_1}, \dots, e^{-\tau x_m}) / \sum_j e^{-\tau x_j}$, and $\mathrm{argmine}$ is the uniform distribution over the set of minimizers.

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Tradeoff with τ (inverse temperature):

- $\tau \uparrow$ implies more concentrated but upper bound $\uparrow$

Mean Field Simulation

- Here we assume perfect simulation of the mean field.
- What if the mean field is *imperfectly* computed?
- For example: obtained by simulating a population of N agents.

  $\Rightarrow$ Extra approximation error (**propagation of chaos**)
- See later: analysis of convergence of Policy Gradient in LQ MFC.

Outline

Motivations for Deep RL in MFMDPs

Issue with Discretization: Curse of Dimensionality
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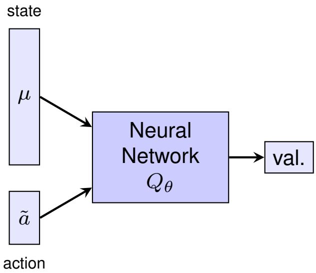
Neural Network as function approximator:

- Replace the Q-table with a network $Q_\theta(\mu, \tilde{a})$.
- Example: **Deep Q-Networks (DQN)**:
  - *Input*: Current distribution μ.
  - *Output*: Level-1 action $\tilde{a}$.



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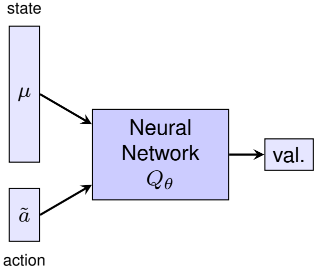

Challenge:

- DQN requires finding $\min_{\tilde{a}} Q_\theta(\mu, \tilde{a})$, which is hard in continuous spaces.
- In MFC, $\tilde{A}$ is often continuous (e.g., the simplex $\mathcal{P}(A)$).

- **Deterministic Policy (Level-1)**:
  - For MFMDP, we have shown that here is an optimal policy which is pure or "deterministic"
  - "Deterministic" refers to the *central planner's choice* of the level-1 action $\tilde{a}$.
  - The individual agents (level-0) can still follow stochastic strategies if $\tilde{a} \in \mathcal{P}(A)^S$.

From DQN to DDPG: Continuous Actions

- **Deterministic Policy (Level-1)**:
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  - "Deterministic" refers to the *central planner’s choice* of the level-1 action $\tilde{a}$.
  - The individual agents (level-0) can still follow stochastic strategies if $\tilde{a} \in \mathcal{P}(A)^S$.
- **Deep Deterministic Policy Gradient (DDPG)**:
  - Designed to handle continuous action spaces directly.
  - Relies on an **Actor** to output the action and a **Critic** to evaluate it.

MF-DDPG: Components and Notation

To train the model, we use the following architectures:

- **Critic Network** $Q_\psi(\mu, \tilde{a})$: Approximates the optimal Q-value (Q^*).
- **Actor Network** $\pi_\phi(\mu)$: Approximates the optimal level-1 policy (π^*).
- **Experience Replay Buffer** $\mathcal{R}$: Stores past transitions $(\mu_n, \tilde{a}_n, \tilde{f}_n, \mu_{n+1})$.
- **Target Networks** $Q_{\tilde{\psi}}, \pi_{\tilde{\phi}}$: Slowly updated to stabilize training.

Algorithm: MF-DDPG

Algorithm: DDPG for MFMDPs

Input: Actor π_ϕ, Critic Q_ψ, Replay Buffer $\mathcal{D}$, episodes N_{total}, length N_{epi}.

Output: Optimized network parameters ϕ^*, ψ^*.

Initialize weights ϕ, ψ; sets targets $\tilde{\phi} \leftarrow \phi, \tilde{\psi} \leftarrow \psi$; initialize $\mathcal{R}$

for $m = 1, \ldots, N_{total}$ **do**

  Sample μ_0

  for $n = 0, \ldots, N_{epi} - 1$ **do**

    Select action $\tilde{a}_n = \pi_\phi(\mu_n) + \epsilon_n^a$, $\epsilon_n^a \sim \mathcal{N}$

    Execute $\tilde{a}_n$, observe cost $\tilde{f}_n$ and new state μ_{n+1}

    Store transition $(\mu_n, \tilde{a}_n, \tilde{f}_n, \mu_{n+1})$ in $\mathcal{R}$

    Sample minibatch from $\mathcal{R}$

    Set target: $y_i = \tilde{f}_i + \gamma Q_{\tilde{\psi}}\big(\mu'_{i+1}, \pi_{\tilde{\phi}}(\mu'_{i+1})\big)$

    Update Critic by minimizing loss: $\mathcal{L}(\psi) = \dfrac{1}{B} \sum_i \big(y_i - Q_\psi(\mu_i, \tilde{a}_i)\big)^2$

    Update Actor by policy gradient:

$$\nabla_\phi J(\phi) \approx \frac{1}{B} \sum_i \nabla_a Q_\psi(\mu_i, a)|_{a=\pi_\phi(\mu_i)} \nabla_\phi \pi_\phi(\mu_i)$$

    Soft update targets: $\tilde{\phi}, \tilde{\psi} \leftarrow \tau(\phi, \psi) + (1 - \tau)(\tilde{\phi}, \tilde{\psi})$

Outline

Overview of Numerical Examples

We illustrate the RL algorithms on two testbeds:

1. **Cybersecurity Model**:
   - $|S| = 4$ and $|A| = 2$.
   - **Goal**: Protect system against external hackers and internal spread.
   - **Benchmark** problem with reference solution.
   - Pure controls at the level-0.
   - **Solvers**: Tabular Q-Learning (Sync/Async) and MF-DDPG.

2. **Discrete Distribution Planning**:
   - Larger state space ($|S| = 11$) and action space ($|A| = 3$).
   - **Goal**: Move population mass to match a target configuration.
   - Mixed controls at the level-0.
   - **Solver**: MF-DDPG only (scales better).

Ex. 1: Cybersecurity – Setup

- **Individual States**: $S = \{DI, DS, UI, US\}$ (Defended/Infected, Susceptible).
- **Planner's Goal**: Protect system against external hackers and internal spread.
- **Transition Matrix** $P^{\mu,a}$:

$$\begin{pmatrix} \dots & P^{\mu,a}_{DS\to DI} & \lambda a & 0 \\ q^D_{rec} & \dots & 0 & \lambda a \\ \lambda a & 0 & \dots & P^{\mu,a}_{US\to UI} \\ 0 & \lambda a & q^U_{rec} & \dots \end{pmatrix}$$

  where infection rates $P_{DS\to DI}$ depend on attack intensity v_H and infected neighbors $\mu(\{DI\}), \mu(\{UI\})$.
- **Time discretization step**: Δt (to approximate continuous-time Markov chains).
- **Social Cost** $\bar{f}(\mu, \bar{a}) = k_D \mu(\{DI, DS\}) + k_I \mu(\{DI, UI\})$.

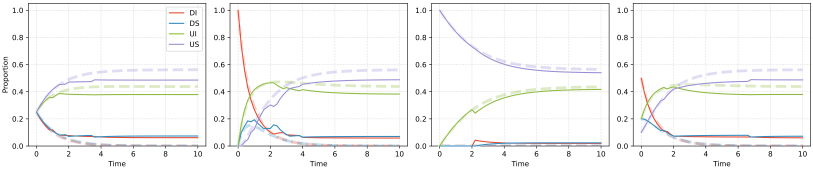
Ex. 1: Cybersecurity – Tabular Q-Learning Solvers

We evaluate three training schemes for the table $\check{Q}(\check{\mu}, \check{a})$:

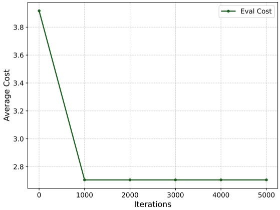
1. **SyncFull**: Updated for all $(\check{\mu}, \check{a})$.
2. **SyncGreedy**: Synchronous update for every state $\check{\mu}$, but using only a single sample for the ϵ-greedy action.
3. **AsyncTraj**: Standard trajectory-based update (Standard Q-learning).

Param	Value	Param	Value
N_m	50	v_H	0.6
Δt	0.2	k_D, k_I	$0.3, 0.5$
T	10.0	q_{rec}^D, q_{rec}^U	$0.5, 0.4$

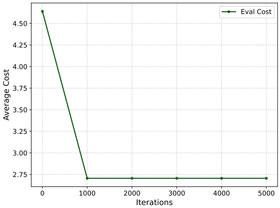
Cybersecurity parameters.



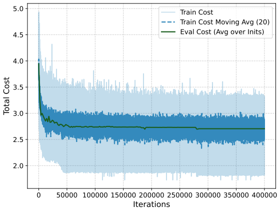
Evaluation trajectories for SyncFull starting from 4 different initializations.



SyncFull



SyncGreedy



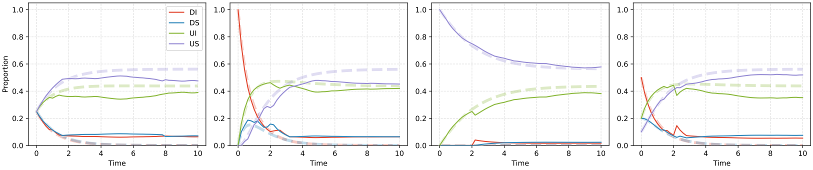
AsyncTraj

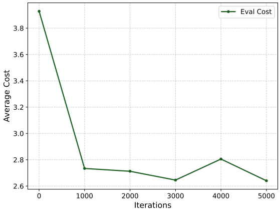
Ex. 1: Cybersecurity – Results: With Common Noise

The attack intensity v_H oscillates according to a stochastic process:

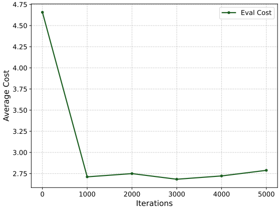
$$v_H^{n+1} = \mathrm{proj}_{[0.7\bar{v}_H, 1.3\bar{v}_H]}(v_H^n + \epsilon_n^0), \quad \epsilon_n^0 \sim \mathcal{N}(0, \sigma^2)$$

with $\bar{v}_H = 0.6$ and $\sigma = 0.05$. Solvers remain robust to these shocks.

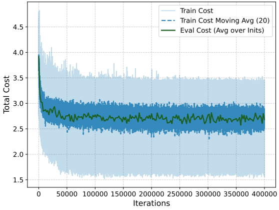




SyncFull



SyncGreedy



AsyncTraj

Ex. 1: Cybersecurity – Value Function

Value function:

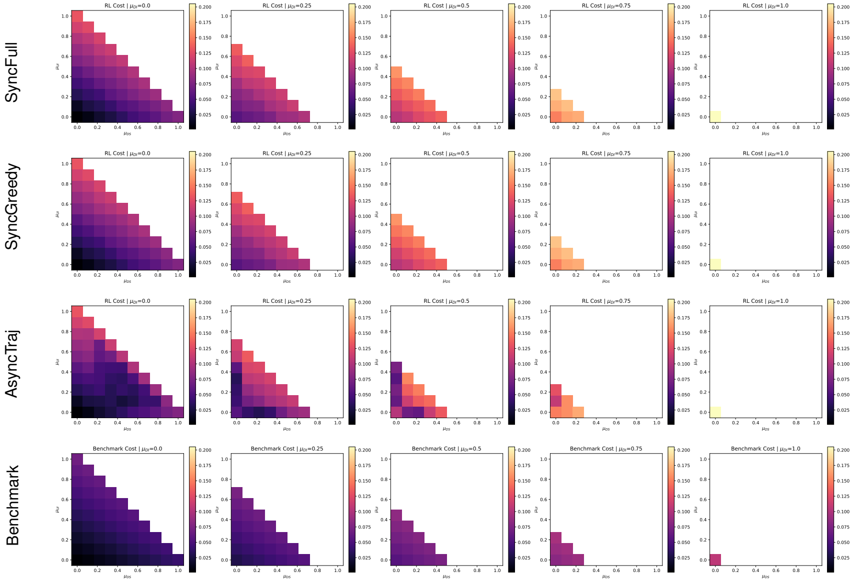
$$\bar{J}^*(\mu) = V^*(\mu) = \min_{\tilde{a}} Q^*(\mu, \tilde{a})$$

is a function of the 4D distribution $\mu \in \mathfrak{S}$.

Visualization:

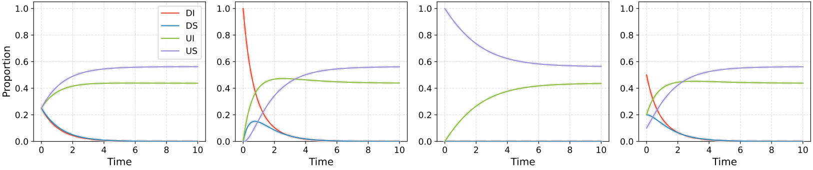
- Define 2D slices by fixing one state, e.g., $\mu_{DI} = \delta \in \{0, 0.25, 0.5, 0.75, 1\}$.
- Vary (μ_{DS}, μ_{UI}) in $[0, 1-\delta]^2$ s.t. $\mu_{DS} + \mu_{UI} \leq 1 - \delta$.
- Remaining mass is $\mu_{US} = 1 - (\mu_{DI} + \mu_{DS} + \mu_{UI})$.
- This allows plotting the 4D simplex on a sequence of 2D triangles.

Ex. 1: Cybersecurity – Tabular Q vs. Benchmark (With CN)

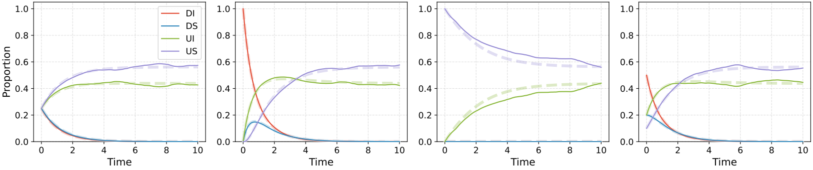


Ex. 1: Cybersecurity – DDPG

- **Continuous Space**: Action space $\mathcal{P}(A)^S \subset [0, 1]^4$ (defense intensity per state).
- **Architecture**: Actor/Critic with 2 layers of 32 neurons.
- **Result**: Trajectories are smoother and closer to benchmark.



DDPG (No CN)



DDPG (With CN)

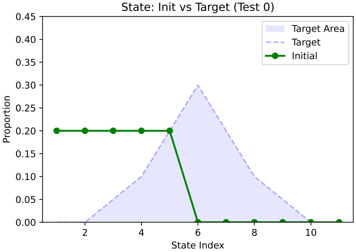
Ex. 2: Discrete Distribution Planning – Setup

Objective: Transport agents to match a target distribution with minimal movement.

- **States**: $S = \{1, \ldots, 11\}$.
- **Actions**: $A = \{-1, 0, 1\}$ (Left, Stay, Right).
- **Dynamics**: $x_{n+1} = \text{proj}_S(x_n + a_n + \varepsilon^0_{n+1})$.
- **Cost Function** (Social Cost):

$$\bar{f}(\mu_n, \tilde{a}_n) = \mathbb{E}_{x \sim \mu_n}[|a_n(x)|] + \lambda \|\mu_n - \mu_{\text{target}}\|_2^2$$

- **Target Profile** μ_{target}: A "Gaussian-like" bump centered at state 6.



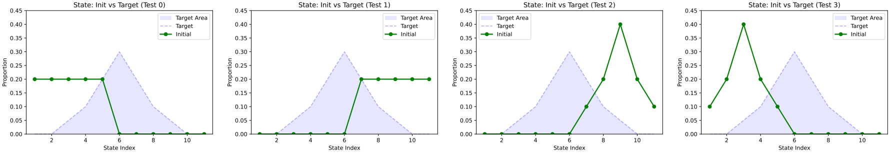
Ex. 2: Discrete Distribution Planning – Setup

Simulation Parameters:

Param	Val	Param	Val
N_{states}	11	T	50
CN Noise	± 1	Prob	0.05
γ	0.95	λ	5.0

DDPG Hyperparameters:

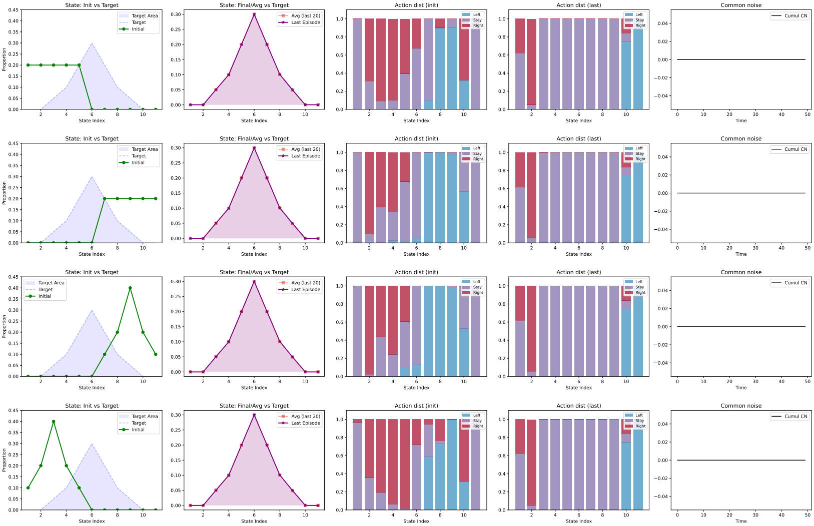
- **Architecture**: MLPs 2 layers, 128 neurons each.
- **Optimizer**: Adam (LR $\approx 10^{-3}$).
- **Target Update**: $\tau \approx 10^{-2}$.
- **Exploration**: Gaussian noise on $\tilde{a}$.



Initial distributions tested in Ex. 2.

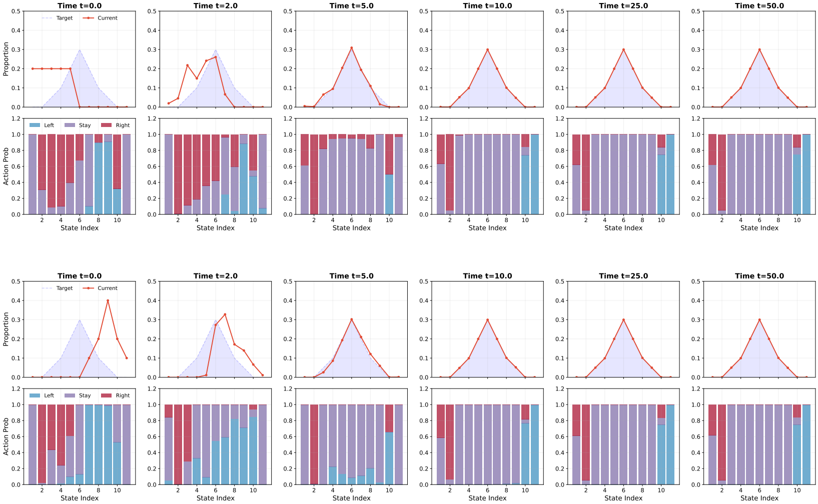
Ex. 2: Discrete Distribution Planning – Results Without Common Noise

Evaluation from 4 different initial distributions (DDPG):



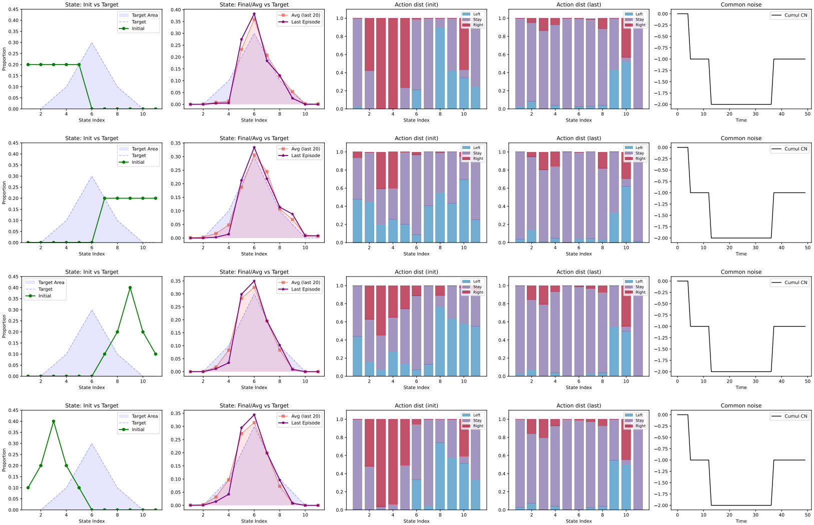
Columns: Initial State, Final State, Initial Action, Terminal Action, Common Noise ($= 0$).

Snapshots of the distribution evolution (no common noise):



Ex. 2: Discrete Distribution Planning – Results With Common Noise

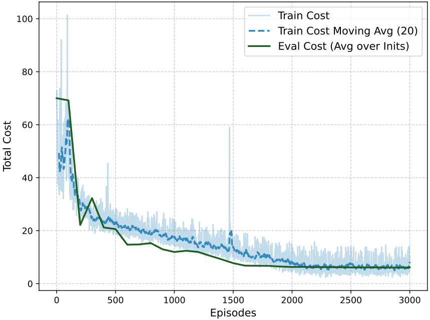
Robustness to stochastic shocks $\varepsilon^0 \in \{-1, 0, 1\}$ (DDPG):



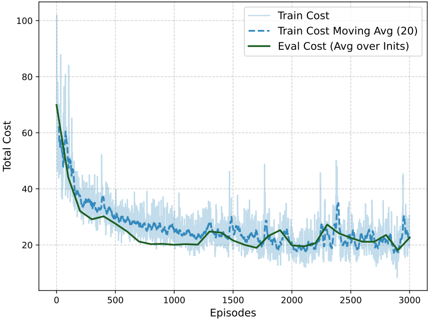
Columns: Initial State, Final State, Initial Action, Terminal Action, Common Noise.

Ex. 2: Discrete Distribution Planning – Learning Progress

DDPG training progress in both deterministic and stochastic environments:



No Common Noise



With Common Noise

- **Efficiency**: DDPG learns effective policies within ∼2000 episodes.
- **Robustness**: Works even with common noise.

Further Reading: MFMDP and MFC Learning

- **MFMDP viewpoint**: population-process MDPs [GG11, GGB12]; common-noise/open-loop MFC formulations [CLT23a, MP22]; dynamic programming and mean field RL [CLT23b, GGWX23, MP23].
- **Tabular Q-learning**: simplex discretization and convergence [CLT19a]; mean field Q-learning variants [GGWX20, CTSK21, AFL22a, AFLZ23].
- **Continuous-time and model-free MFC**: continuous-time Q-learning [WY25]; model-free policy gradient and actor-critic methods [MPR26, BK25, FGL+25, PW25].
- **Model-based, decentralized, robust, safe**: model-based MF-MARL [PKB23a, PKB23b, HYH24, HHK24]; decentralized network approaches [GGWX21a, GGWX24]; robust/social-control and safe mean-field MARL [XCW+25, JPJ+24].

Outline

This part is mostly based on:

Carmona, Lauriere, Tan: arXiv:1910.04295

and lecture notes

Carmona & Lauriere: Mean Field Reinforcement Learning
(arXiv:2607.01525)

Outline

LQMFC: Model and Notations

Representative Agent Dynamics: For $n \geq 0$:

$$x_{n+1} = Ax_n + \bar{A}\bar{x}_n + Bu_n + \bar{B}\bar{u}_n + \varepsilon^0_{n+1} + \varepsilon_{n+1}$$

where:

- $x_n \in \mathbb{R}^d$: state, $\quad u_n \in \mathbb{R}^\ell$: action.
- $\bar{x}_n = \mathbb{E}[x_n|\mathcal{F}^0_n]$ and $\bar{u}_n = \mathbb{E}[u_n|\mathcal{F}^0_n]$: conditional means.
- ε^0_{n+1} (common) and ε_{n+1} (idiosyncratic) noise terms.

Mean Field Cost: for $\mathbf{u} = (u_n)_{n\geq 0}$

$$J(\mathbf{u}) = \mathbb{E}\left[\sum_{n\geq 0} \gamma^n f(x_n, \bar{x}_n, u_n, \bar{u}_n)\right]$$

where f is of “tracking” form:

$$\begin{aligned} f(x, \bar{x}, u, \bar{u}) = (x - \bar{x})^\top Q(x - \bar{x}) + \bar{x}^\top (Q + \bar{Q})\bar{x} \\ + (u - \bar{u})^\top R(u - \bar{u}) + \bar{u}^\top (R + \bar{R})\bar{u}. \end{aligned}$$

Optimal Control Expression

Theorem (Optimal Control)

The optimal control u_n^* is in feedback form and linear in $(x_n, \bar{x}_n)$:

$$u_n^* = -K^*(x_n - \bar{x}_n) - L^*\bar{x}_n$$

where the optimal parameters are:

$$K^* = \frac{1}{2}R^{-1}B^\top P, \quad L^* = \frac{1}{2}(R+\bar{R})^{-1}(B+\bar{B})^\top \bar{P}$$

The matrices $P, \bar{P} \in \mathbb{R}^{d\times d}$ satisfy the discrete-time Riccati equations:

$$P = \gamma(A^\top P + 2Q)(A - BR^{-1}B^\top P/2)$$
$$\bar{P} = \gamma(\tilde{A}^\top \bar{P} + 2\tilde{Q})(\tilde{A} - \tilde{B}\tilde{R}^{-1}\tilde{B}^\top \bar{P}/2)$$

with $\tilde{A} = A + \bar{A}$, $\tilde{B} = B + \bar{B}$, $\tilde{Q} = Q + \bar{Q}$, $\tilde{R} = R + \bar{R}$.

From MFC to N Agents

Once the optimal gains (K^*, L^*) are known, how does agent i apply the control?

1. If the MFC mean $\bar{x}_n$ is known for all n (e.g., fixed initial condition and no common noise):
$$u_n^i = -K^*(x_n^i - \bar{x}_n) - L^*\bar{x}_n$$
The control is **decentralized**: agent i only needs to observe x_n^i.

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2. If $\bar{x}_n$ is unknown (e.g., due to common noise):
   $$u_n^i = -K^*(x_n^i - \hat{x}_n^N) - L^*\hat{x}_n^N, \quad \text{with } \hat{x}_n^N = \frac{1}{N}\sum_{j=1}^{N} x_n^j$$
   The control is **not fully decentralized** as it requires the empirical mean $\hat{x}_n^N$. However, in the LQMFC case, it only requires the mean, not the full distribution.

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Note: In general MFC (non-LQ), optimal control might depend on the **full distribution** in complex ways, not just the mean.

Cost Decomposition

Idea: Since the optimal control is linear, use Policy Gradient (PG) to search directly for $\theta = (K, L) \in \Theta$.

$$C(\theta) := J(\mathbf{u}^{\theta}).$$

Cost Decomposition

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$$C(\theta) := J(\mathbf{u}^\theta).$$

The cost $C(\theta)$ and decomposes as:

$$C(\theta) = C_y(K) + C_z(L)$$

where C_y (idiosyncratic) and C_z (aggregate) are standard LQR costs:

$$C_y(K) = \mathbb{E}\left[\sum_{n\geq 0} \gamma^n (y_n^\top (Q + K^\top RK) y_n)\right] = \mathrm{Tr}((Q + K^\top RK)\Sigma_K)$$

$$C_z(L) = \mathbb{E}\left[\sum_{n\geq 0} \gamma^n (z_n^\top (\tilde{Q} + L^\top \tilde{R}L) z_n)\right] = \mathrm{Tr}((\tilde{Q} + L^\top \tilde{R}L)\Sigma_L)$$

where the **state variance matrices** are defined as:

$$\Sigma_K = \mathbb{E}\left[\sum_{n\geq 0} \gamma^n y_n y_n^\top\right], \quad \Sigma_L = \mathbb{E}\left[\sum_{n\geq 0} \gamma^n z_n z_n^\top\right]$$

Gradient Expression

Theorem (Policy Gradient Expression)

For $\theta = (K, L) \in \Theta$, the gradient of the cost $C(\theta)$ is:

$$\nabla_K C(\theta) = 2E_K \Sigma_K, \quad \nabla_L C(\theta) = 2E_L \Sigma_L$$

where $E_K = (R + \gamma B^\top P_K^y B)K - \gamma B^\top P_K^y A$, and Σ_K is the state variance matrix.

Remarks:

- Gradient depends on the state variance matrix Σ_K and E_K, which depend on the model parameters. This is not model-free.
- Mirror results for standard LQR (Fazel et al., 2018).
- Fundamental for both analytical convergence and model-free implementations.
- Set of admissible policies: $\Theta :=$ $\left\{\theta = (K, L) \in \mathbb{R}^{\ell \times d} \times \mathbb{R}^{\ell \times d} \mid \gamma \|\mathrm{A} - \mathrm{B}K\|^2 < 1,\ \gamma \|\mathrm{A} + \bar{\mathrm{A}} - (\mathrm{B} + \bar{\mathrm{B}})L\|^2 < 1\right\}$

Outline

Exact Policy Gradient Algorithm

Setup: Full knowledge of model parameters $(A, \bar{A}, B, \bar{B}, Q, \bar{Q}, R, \bar{R})$.

Algorithm: Exact Policy Gradient for MFC

Input: Initial policy $\theta^{(0)} = (K^{(0)}, L^{(0)}) \in \Theta$, learning rate η.
Output: Optimal policy θ^*.

for $k = 0, 1, 2, \dots$ **do**
    Compute $\nabla_K C(K^{(k)})$ and $\nabla_L C(L^{(k)})$
    Update $K^{(k+1)} \leftarrow K^{(k)} - \eta \nabla_K C(K^{(k)})$
    Update $L^{(k+1)} \leftarrow L^{(k)} - \eta \nabla_L C(L^{(k)})$

The convergence analysis relies on two key properties of the cost functions:

1. Polyak-Lojasiewicz (PL) Condition:

$$\|\nabla C(\theta)\|^2 \geq \nu_{pl}(C(\theta) - C(\theta^*))$$

Even though the cost is non-convex, the gradient norm bounds the optimality gap.

2. Local Smoothness: The cost is smooth in the direction of the gradient descent for small enough steps:

$$C(\theta^{(k+1)}) - C(\theta^{(k)}) \leq \langle \nabla C(\theta^{(k)}), \theta^{(k+1)} - \theta^{(k)} \rangle + \frac{L_{loc}}{2} \|\theta^{(k+1)} - \theta^{(k)}\|^2$$

Global Convergence of Exact PG

Theorem (Convergence of Exact PG)

*Under standard assumptions, if the learning rate $\eta > 0$ is sufficiently small, the costs converge **linearly** to the optimum:*

$$C(\theta^{(k)}) - C(\theta^*) \leq (1 - \eta\nu_{pl})^k (C(\theta^{(0)}) - C(\theta^*))$$

where $\nu_{pl} > 0$ is the PL constant.

Corollary (Iteration Complexity)

To reach a precision $\varepsilon > 0$, i.e., $C(\theta^{(k)}) - C(\theta^*) \leq \varepsilon$, it is sufficient to take:

$$k \geq \frac{1}{\eta\nu_{pl}} \log\left(\frac{C(\theta^{(0)}) - C(\theta^*)}{\varepsilon}\right)$$

Sketch of Proof

1. **Admissibility**: Prove that for η small enough, the updated parameter $\theta^{(k+1)}$ remains in the set of stabilizing policies Θ.

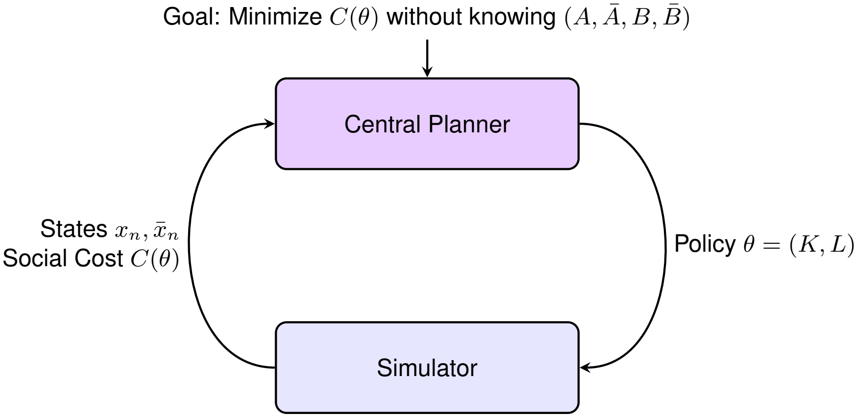
2. **Descent Guarantee**: Use **local smoothness** to bound the decrease in cost at each step:
$$C(\theta^{(k+1)}) - C(\theta^{(k)}) \leq -\eta(1 - \text{bias})\|\nabla C(\theta^{(k)})\|^2$$

3. **Optimality Gap**: Invoke the **PL condition** to link the gradient norm to the distance from the global minimum:
$$C(\theta^{(k+1)}) - C(\theta^{(k)}) \leq -\eta\nu_{pl}(C(\theta^{(k)}) - C(\theta^*))$$

4. **Iteration**: Re-arrange terms to show the geometric decay of the optimality gap.

What about model-free PG?

The LQMFC Reinforcement Learning Paradigm



Challenge: We only have access to **cost evaluations** based on **trajectories**.
Simulator: MKV-based or (finite) Population-based.

Outline

MKV Simulator $\mathcal{S}^T_{MKV}$:

- Fix policy $\theta = (K, L)$
- Fix number of time steps T
- Simulate a trajectory following the McKean-Vlasov (= mean field) dynamics
- Return the truncated cost $\tilde{C}^T(\theta) = \sum_{n=0}^{T-1} \gamma^t c_n(\theta)$

Model-Free MKV-Based PG

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Zero-th order optimization:

- Estimate the gradient based only on samples of the cost;
- Plug this estimate in the PG algorithm instead of the true gradient.
- Uses $2M$ samples uniformly distributed on the unit sphere of radius τ.

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Zero-th order optimization:

- Estimate the gradient based only on samples of the cost;
- Plug this estimate in the PG algorithm instead of the true gradient.
- Uses $2M$ samples uniformly distributed on the unit sphere of radius τ.

Algorithm: MKV-Based Gradient Estimation

Input: Parameter $\theta = (K, L)$, horizon T, M perturbations, radius τ.
Output: $\tilde{\nabla}^{T,M,\tau}(\theta)$.

1 **for** $i = 1, \dots, M$ **do**
2 Sample $v_i^{(idy)}, v_i^{(com)} \sim$ Uniform$(\mathbb{S}_\tau)$
3 Set $\theta_i = (K + v_i^{(idy)}, L + v_i^{(com)})$
4 Receive $\tilde{C}^T(\theta_i)$ from $\mathcal{S}^T_{MKV}(\theta_i)$
5 Return $\tilde{\nabla}_K^{T,M,\tau}(\theta) = \dfrac{\ell d}{\tau^2} \dfrac{1}{M} \displaystyle\sum_{i=1}^{M} \tilde{C}^T(\theta_i) v_i^{(idy)}, \tilde{\nabla}_L^{T,M,\tau}(\theta) = \dfrac{\ell d}{\tau^2} \dfrac{1}{M} \displaystyle\sum_{i=1}^{M} \tilde{C}^T(\theta_i) v_i^{(com)}$.

Intuition for Zeroth-Order Optimization

Idea: Estimate the gradient of a function f by taking a random unit vector v and scaling it by $f(x + \tau v)$. The expectation is proportional to the gradient of a *smoothed* version of f.

Lemma 2.1 of Flaxman et al. (2005)

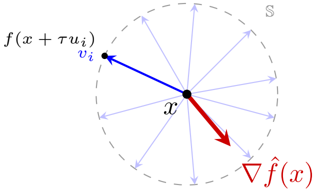
The expectation over the sphere is proportional to the smoothed gradient:

$$\mathbb{E}_{v \in \mathbb{S}}[f(x + \tau v)v] = \frac{\tau}{d} \nabla \hat{f}(x)$$

where $\hat{f}$ is the smoothed function defined as:

$$\hat{f}(x) = \mathbb{E}_{u \in \mathbb{B}}[f(x + \tau u)]$$

where $\mathbb{B}$ is the unit ball in $\mathbb{R}^d$ and u is uniform over the ball.



Intuition: Sampling on the sphere $\mathbb{S}_\tau$ gives the direction of the gradient of the smoothed function $\hat{f}$.

Global Convergence of PG with MKV Simulator

Theorem (Convergence with MKV Simulator)

For a target precision $\varepsilon > 0$, if η is small enough and (T, M, τ) satisfy:

- $\tau \lesssim \varepsilon$ *(Small perturbation radius)*
- $T \gtrsim \log(1/\varepsilon)$ *(Sufficiently long horizon)*
- $M \gtrsim$ *poly*(d, ℓ, δ^{-1}) *(High-probability concentration)*

Then with probability $\geq 1 - \delta$, *after* $k \geq \frac{2}{\eta \nu_{pl}} \log(\frac{C_0}{\varepsilon})$ *steps, we have:*

$$C(\theta^{(k)}) - C(\theta^*) \leq \varepsilon$$

Note: We recover **linear convergence** up to the precision level ε despite using only noisy cost evaluations from the MKV simulator.

Sketch of Proof: Error Decomposition

The proof decomposes the error $\|\tilde{\nabla} - \nabla C\|$ into three terms:

1. **Smoothing Error**: Bias introduced by estimating the gradient of a smoothed version of the cost
   - Controlled by τ.
2. **Truncation Error**: Bias from approximating the infinite-horizon cost $C(\theta)$ with a finite-horizon sum $\tilde{C}^T$
   - Controlled by $\gamma^T \to 0$ as $T \to +\infty$.
3. **Sampling Error**: Variance from using M samples and stochastic noise in the MKV simulation
   - Controlled by M via concentration inequalities.

Combine these with the **PL condition** and **local smoothness** to show that the noisy update still follows the direction of descent.

Outline

Algorithm: Population-Based PG

Population Simulator $\mathcal{S}_{pop}^{T,N}$: Simulates N agents following the same policy θ.

$$\tilde{C}^{T,N}(\theta) = \sum_{n=0}^{T-1} \gamma^n \left(\frac{1}{N} \sum_{j=1}^{N} f(X_n^{(j),\theta}, \bar{X}_n^{N,\theta}, u_n^{(j),\theta}, \bar{u}_n^{N,\theta}) \right)$$

Sources of error:

- trajectory truncation: T
- zero-order gradient estimation: τ, M
- mean field approximation by finite population: N

Algorithm: Population-Based PG

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- zero-order gradient estimation: τ, M
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Algorithm: Population-Based Gradient Estimation

Input: Parameter $\theta = (K, L)$, horizon T, population size N, M perturbations, radius τ.

Output: $\tilde{\nabla}^{T,N,M,\tau}(\theta)$.

for $i = 1, \ldots, M$ **do**

    Sample $v_i^{(idy)}, v_i^{(com)} \sim$ Uniform$(\mathbb{S}_\tau)$

    Set $\theta_i = (K + v_i^{(idy)}, L + v_i^{(com)})$

    Receive $\tilde{C}^{T,N}(\theta_i)$ from $\mathcal{S}_{pop}^{T,N}(\theta_i)$

Return gradients using zeroth-order formula as in MKV case.

Theorem (Convergence with Population Simulator)

If the population size N is sufficiently large, specifically:

$$N \geq \mathcal{O}\left(\frac{C_0}{\varepsilon \eta \nu_{pl}}\right)$$

and hyperparameters (τ, T, M) are tuned as in the MKV case, then with high probability:

$$C(\theta^{(k)}) - C(\theta^*) \leq \varepsilon$$

Consequence: A finite population can learn a policy that is approx. optimal for the **infinite-population** MF limit.

Theory: Population Approximation

Key step: Approximation of the MF behavior by N agents.

Variance Identities: The noise and initial variances for the N-agent system are shifted by $1/N$:

$$\Sigma^{1,N} = (1 - 1/N)\Sigma^1, \qquad \Sigma^{0,N} = \Sigma^0 + \Sigma^1/N$$

$$\Sigma_{y_0}^N = (1 - 1/N)\Sigma_{y_0}, \qquad \Sigma_{z_0}^N = \Sigma_{z_0} + \Sigma_{y_0}/N$$

Cost Difference: The difference between the social cost C^N and the MF cost C scales as $O(1/N)$. More precisely (see paper / lecture notes for notations):

$$|C^N(\theta) - C(\theta)| = \left| \frac{1}{N} \left\langle P_L^z - P_K^y, \, \Sigma_{y_0} + \frac{\gamma}{1-\gamma}\Sigma^1 \right\rangle_{tr} \right|$$

which leads to the bound:

$$|C^N(\theta) - C(\theta)| \leq \frac{1}{N}(\|P_K^y\| + \|P_L^z\|)\frac{\max\{Tr(\Sigma_{y_o}), Tr(\Sigma^1)\}}{1-\gamma}$$

Hence: The bias introduced by the finite population disappears as $N \to \infty$.

General Case: Conditional Propagation of Chaos

See Carmona and Delarue (2018), Volume II, for more details.

Theorem 2.12 *Within the above framework and under assumption* **Conditional MKV SDE**, *the system of particles* (2.3) *with* $(X_0^1, \cdots, X_0^N)$ *as initial condition has a unique solution for every* $N \geq 1$. *It is denoted by* $((X_t^{N,i})_{0 \leq t \leq T})_{i=1,\cdots,N}$. *Moreover,*

$$\lim_{N \to \infty} \left(\max_{1 \leq i \leq N} \mathbb{E}\Big[\sup_{0 \leq t \leq T} |X_t^{N,i} - \underline{X}_t^i|^2 \Big] + \sup_{0 \leq t \leq T} \mathbb{E}\Big[W_2\Big(\frac{1}{N} \sum_{i=1}^N \delta_{X_t^{N,i}}, \mathcal{L}^1(\underline{X}_t^1) \Big)^2 \Big] \right) = 0.$$

When $\mathbb{E}^1[|X_0^1|^q] < \infty$ *for some* $q > 4$, *there exists a constant* C, *only depending on* T, $\mathbb{E}^1[|X_0^1|^q]$ *and the Lipschitz constants of* b, σ *and* σ^0, *such that:*

$$\max_{1 \leq i \leq N} \mathbb{E}\Big[\sup_{0 \leq t \leq T} |X_t^{N,i} - \underline{X}_t^i|^2 \Big] + \sup_{0 \leq t \leq T} \mathbb{E}\Big[W_2\Big(\frac{1}{N} \sum_{i=1}^N \delta_{X_t^{N,i}}, \mathcal{L}^1(\underline{X}_t^1) \Big)^2 \Big] \leq C\epsilon_N,$$

where $(\epsilon_N)_{N \geq 1}$ *satisfies:*

$$\epsilon_N = \begin{cases} N^{-1/2}, & \text{if } d < 4, \\ N^{-1/2} \log N, & \text{if } d = 4, \\ N^{-2/d}, & \text{if } d > 4. \end{cases} \tag{2.7}$$

Outline

Experimental Setup

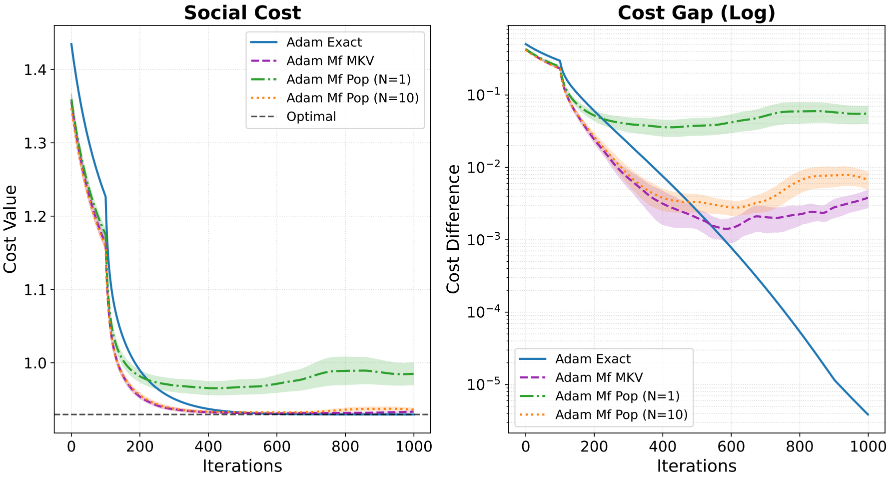
Model: Scalar LQ-MFC with idiosyncratic and common noise.

- **Learner**: Central planner using Adam for parameter updates.
- **Comparison**:
  - Exact PG
  - MF-MKV
  - MF-Pop with $N = 1$ and 10

Parameter	Value	Hyperparameter	Value
Dynamics $(a, \bar{a})$	$(0.5, 0.5)$	Learning rate η	0.01
Control $(b, \bar{b})$	$(0.5, 0.5)$	Radius τ	0.1
State cost $(q, \bar{q})$	$(0.5, 0.5)$	Horizon T	10
Control cost $(r, \bar{r})$	$(0.5, 0.5)$	Directions M	100
Discount factor γ	0.9	Iterations	1000
Variances σ^2	0.01	Seeds	10

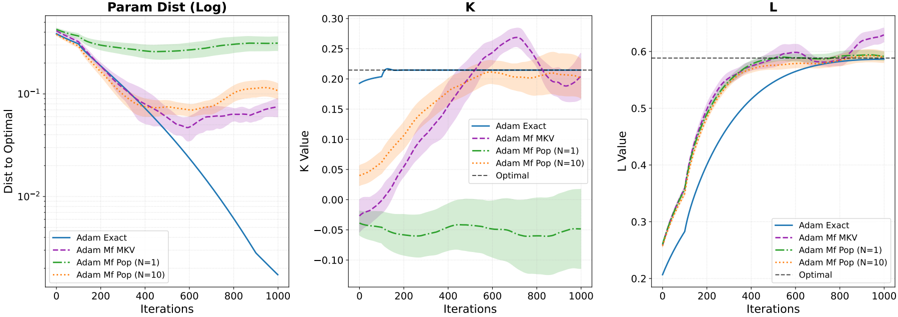
Learning rate: For practical efficiency and stability, we adapt the learning rate $\eta = \eta^{(k)}$ during iterations, using Adam optimizer.

Numerical Results: Cost Convergence



- Convergence to the theoretical optimal cost (black dashed line).
- **MKV simulator** (Purple dashed) closely follows the **Exact gradient** (Blue).
- **Pop. simulator**: High var. and bad convergence for $N = 1$; improve for $N = 10$.

Numerical Results: Parameter Convergence



- Convergence of parameters (K, L) toward (K^*, L^*).
- Left: Distance to optimum. Middle/Right: K/L.
- Stable convergence across all model-free estimators.
- But for $N = 1$, K is not learned... and cannot be learned.

Further Reading: Policy Optimization for LQ MFC

- **LQ mean field control**: policy-gradient analyses [CLT19b, CLT19c, WHYW21a].
- **Actor–critic and model-free variants**: LQ MFG actor–critic and PG [FYCW19a, FYCW19b, uZZMB20a, WHYW21b]; model-free social-control learning [WZZ20, LLX24].
- **Social optimization and robustness**: LQG social optimum by RL [XWS25]; robust mean field social control [XCW+25]; zero-sum mean-field type games [CHLT20a, CHLT21a, CHLT20b, CHLT21b].
- **Recent PG directions**: model-free PG beyond LQ structure [MPR26, FGL+25]; actor–critic with moment neural networks [PW25]; independent RL for cooperative–competitive agents [uZKLB24a].

Outline

This part is mostly based on:

"Learning in Mean Field Games: A Survey",
Lauriere, Perrin, Perolat, Girgin, Muller, Elie, Geist, Pietquin
(arXiv:2205.12944)

MFG Operators and Equilibrium

Two key operators define the Mean Field Game dynamics:

- **Mean Field Operator (MF)**: Maps a policy π to the sequence of distributions $\mu = \text{MF}(\pi)$ it generates.
- **Best Response Operator (BR)**: Maps a distribution flow μ to the set of policies $\pi \in \text{BR}(\mu)$ that minimize the individual cost $J(\pi; \mu)$.

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MFG Nash Equilibrium (MFNE)

A pair $(\hat{\pi}, \hat{\mu})$ is an MFNE if and only if $\hat{\pi} \in \text{BR}(\hat{\mu})$ and $\hat{\mu} = \text{MF}(\hat{\pi})$.

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A pair $(\hat{\pi}, \hat{\mu})$ is an MFNE if and only if $\hat{\pi} \in \text{BR}(\hat{\mu})$ and $\hat{\mu} = \text{MF}(\hat{\pi})$.
This can be rewritten as a fixed point over policies or distributions:

$$\hat{\pi} \in \text{BR}(\text{MF}(\hat{\pi}))$$

or (if the BR is unique)

$$\hat{\mu} = \text{MF}(\text{BR}(\hat{\mu}))$$

Iterative Methods for MFG

Based on the fixed point formulation, we can define iterative algorithms.

- **Fixed Point Iteration**: Alternates directly between BR and MF updates:

$$\pi_{k+1} \in \mathrm{BR}(\mu_k)$$
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  - *Warning*: Often fails to converge due to oscillations. BR operator $\mathrm{BR}(\mu)$ might not be unique.

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  - *Warning*: Often fails to converge due to oscillations. BR operator $\text{BR}(\mu)$ might not be unique.

- **Policy Iteration for MFG**: Intertwines standard MDP Policy Evaluation and Improvement with MF updates:

$$\begin{aligned} Q_{k+1} &= Q^{\mu_k,\pi_k} && \text{(Policy Evaluation)} \\ \pi_{k+1} &\in \arg\min Q_{k+1} && \text{(Policy Improvement)} \\ \mu_{k+1} &= \text{MF}(\pi_{k+1}) && \text{(Mean Field Update)} \end{aligned}$$

Fixing Oscillations: Regularization and Averaging

To prevent oscillations and ensure uniqueness of the Best Response, we use two main techniques:

1. **Entropic Regularization**: Modify the cost by adding an entropy penalty $\eta \log(\pi(a|x))$. This makes the optimal policy a softmin policy (smooth), ensuring uniqueness. Tuning the temperature gives smoothness.

2. **Averaging Iterations**: Do not jump directly to the new BR/MF. Average the history of past strategies or distributions.
   - **Fictitious Play (FP)**: Averages policies or distributions over iterations.
   - **Online Mirror Descent (OMD)**: Maintains a cumulative Q-function to softly update the policy.

Fictitious Play (FP)

Averages past distributions.

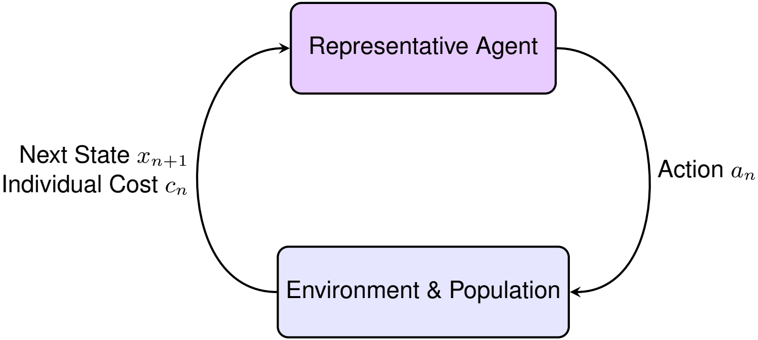
1. $\pi_{k+1} \in \mathsf{BR}(\bar{\mu}_k)$
2. $\mu_{k+1} = \mathsf{MF}(\pi_{k+1})$
3. $\bar{\mu}_{k+1} = \frac{k}{k+1}\bar{\mu}_k + \frac{1}{k+1}\mu_{k+1}$

Online Mirror Descent (OMD)

Accumulates evaluating Q-functions.

1. $Q_{k+1} = Q^{\mu_k,\pi_k}$
2. $\bar{Q}_{k+1} = \bar{Q}_k + \alpha Q_{k+1}$
3. $\pi_{k+1} \propto \exp(-\bar{Q}_{k+1}/\tau)$
4. $\mu_{k+1} = \mathsf{MF}(\pi_{k+1})$

The MFG RL Paradigm



The population mean field μ evolves internally. The agent **only** observes its individual state and cost.

MFG in the Model-Free RL Setting

How do we implement the fixed point iterations without knowing the model?

- **Inner Loop (RL)**: Since μ is fixed (or $\bar{\mu}$ in FP), the problem is a standard MDP.
  - We can compute the **Best Response** using standard model-free RL algorithms (Q-learning, DQN, PPO, DDPG) running in a simulated environment with frozen population behavior.
  - For **Policy Iteration / OMD**, we replace the exact evaluator Q^{μ_k,π_k} with a model-free **Policy Evaluation** step (e.g., TD-learning).

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- **Outer Loop (MF Update)**:
  - Once the inner loop RL converges, we deploy the new policy π_{k+1} (MKV simulator or Population simulator).
  - We obtain the new mean field μ_{k+1}.
  - Apply the FP averaging or OMD accumulation, and repeat the outer loop.

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  - Apply the FP averaging or OMD accumulation, and repeat the outer loop.

- **Mixture:** the two loops can be "mixed" by adjusting the number of iterations in each loop.

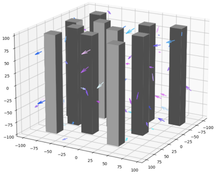
MFG Numerical Example: Flocking

Setting [PLP+21]:

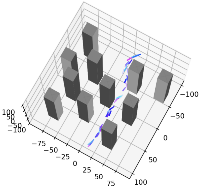
- Population of birds adjusting their acceleration to match the average velocity of their neighbors.
- Continuous state space: positions and velocities. Continuous action space: acceleration.
- **Model-Free Approach**: Since state/action spaces are continuous, we cannot use tabular methods.

Algorithm: Deep Fictitious Play

- **Distribution** μ: Modeled with a Normalizing Flow.
- **Policy** π: Learned via Deep RL (Soft Actor-Critic – SAC).



Initial distribution



At convergence

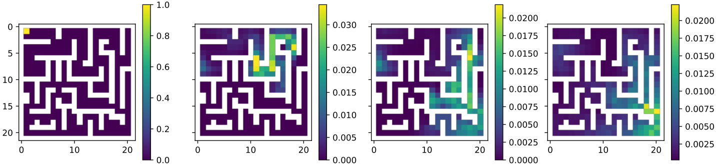
MFG Numerical Example: Crowd Exiting a Maze

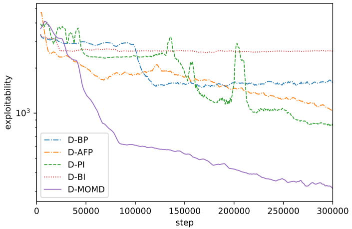
Setting [LPG+22]:

- A large crowd of agents tries to exit a maze while avoiding congested areas.
- Demonstrates the scalability of advanced algorithms for Mean Field Games.

Algorithm: Deep Online Mirror Descent (OMD) and Deep Fictitious Play (FP)

- Uses Neural Networks to approximate the cumulative Q-function and the population distribution.





Resources

More details:

- **Papers**:
  - "Learning in Mean Field Games: A Survey", Lauriere et al. (arXiv:2205.12944)
  - References at the end
- **Personal webpage**:
  - slides
  - codes

Further Reading: Learning in MFG

- **Surveys, software, benchmarks**: learning MFG survey [LPP+]; scalable deep RL algorithms and OpenSpiel examples [LPG+22, LLL+19]; Bench-MFG [MSG+26].
- **Fixed-point, value, and Q-learning**: value iteration and Q-learning [GHXZ19, AKS20, AKS19, AKS23, AKS21, GHXZ23].
- **Fictitious play, OMD, policy iteration**: learning and averaging schemes [PPL+20, PPE+21a, CCG21, CT21, LST21].
- **Deep RL, MARL, inverse RL**: deep RL and distribution embeddings [EPL+20, CK21a, PLP+21, MSW+25]; mean-field MARL and inverse RL [YLL+18, YYT+17, CLK21, CZLH22, RKP+24, ZKBB23].

Outline

Summary

- **Mean Field RL Paradigm**: Central planner viewpoint for cooperative agents using the Mean Field Markov Decision Process (MFMDP) framework.
- **Discrete Settings**:
  - **Model-Free Q-learning**: Tabular approach with convergence guarantees.
  - **Exploration/Exploitation**: Sweep-based vs trajectory-based updates.
- **Continuous and Deep RL**:
  - **MF-DDPG**: Scaling to continuous state-action spaces using deep NN.
  - **Scalability**: Handling complex environments.
- **Linear-Quadratic MFC**:
  - **Global Convergence**: Proofs for Policy Gradient (PG) methods.
  - Decoupling of idiosyncratic and aggregate states via PL condition.
- **Practical Implementation**:
  - Model-free PG using zeroth-order estimation (MKV and Pop. simulators).
  - Robustness of Adam-based updates in stochastic environments.

Extensions and Variants

- **Problem Settings**:
  - Finite horizon, ergodic/average cost, ...
  - Entropy-regularized, risk-sensitive objectives ...
- **Beyond Homogeneity and Symmetry**:
  - Survey: An overview of some extensions of mean field games beyond perfect homogeneity and anonymity, L., Science China Information Sciences (2026)
  - Multi-population models: several types of agents.
  - Graphon control: non-exchangeable agents on large graphs.
  - ...
- **Information Structure**:
  - Class of policies: centralized vs decentralized.
  - Partial observation, imperfect information, ...
- **Beyond Pure Cooperation**:
  - MFG: Non-cooperative settings (Nash equilibria).
  - MF Type Games / MF Teams: Nash eq. between cooperative teams.
  - MF Control Games: Limit of infinitely many competing teams.
  - Mixed Individual/Population Games
  - ...

Future Directions

- **Theoretical RL**:
  - Theoretical convergence for Deep RL for MFC?
  - Rates of convergence?
- **Mean Field Representation**:
  - Efficient approximation of probability measures?
  - How does it interact with the RL algorithm?
- **Extensions of MFC**:
  - RL for the variants mentioned in the previous slide?
- **Real-World Scenarios**:
  - Concrete applications?
  - Data-driven approaches?
  - Ongoing progress:
    - A game-theoretic framework for generic second-order traffic flow models using mean field games and adversarial inverse reinforcement learning, Mo et al., Transportation Science (2024)
    - Inverse Problems for Costs and Controls in LQG MFGs via Mean Field Trajectories, Lambrecht & Lauriere, CDC'26
    - ...

Further Reading: Broader Directions

- **Multiple populations and heterogeneity**: multi-population/team learning [SPTH20, SSM18, GSPTH20, PPE+22, YVCD20, uZMB23]; graphon learning [CK21b, FCK22, CKK22, FCK23, HWYZ23, ZTWY24].
- **Beyond pure cooperation**: mean field type/control games and mixed games [ADF+22, ADF+23, CDDL23]; zero-sum settings [CHLT20b, CHLT21b, GAT23, uZKLB24b].
- **Correlated and equilibrium learning**: policy-space response oracles and correlated equilibria [MRE+22, MER+22, WCL+22].
- **Partial observation**: partially observed MFC/MFG and decentralized policies [CHFK23, CHFK24b, STCP20, CHFK24a, YLZW23].

Further Reading: Data, Robustness, Benchmarks

- **Model-based and sample-efficient RL**: model-based MF-MARL [PKB23b, HYH24, HHK24]; decentralized learning [YAY22, STCP22, YCGH23, YGH24].
- **Inverse and imitation learning**: inverse RL for MFG/MFC [YYT+18, CLK21, CZLH22, RKP+24, AKS24]; traffic-flow inverse RL [MCD+24].
- **Representation and diagnostics**: distribution approximation in flocking and continuous MFGs [PLP+21, MSW+25]; OpenSpiel and Bench-MFG environments [LLL+19, MSG+26].
- **Regularization, exploration, and safety**: exploration noise [DV21]; regularized/stable learning [ASA+23, MQW+22]; robust and safe social control [XCW+25, JPJ+24].

Thank you!

Slides:

https://mlauriere.github.io/slides-public/2026-changsha-mfrl-tutorial.pdf

Feel free to reach out if you have questions: mathieu.lauriere@nyu.edu

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